



Information Systems for Big Data

Prof. G. Fenza gfenza@unisa.it

Microsoft Teams



- Join team by code: **wlm3kcx**

1st Lesson – Roadmap



- About
 - Me
 - Lessons Plan
 - Exam



Di cosa mi occupo



- 2005-2008

- Ontologies
- Semantic Search
- Semantic Web Services
- Fuzzy Data Analysis

Utilizzo dei modelli
semantici heavy

- 2008-2013

- Text Mining / NLP
- Semantic Api
- Question Answering
- Ontology Learning

Definizione di metodi
per estrarre modelli
semantici light-weight da
testo

- 2013-2017

- Speech/Video Analysis
- Social Media Analytics
- SEO, Ad Placement

Startupper, estrazione
conoscenza anche da
contenuti multimediali
e social media



SERICS

- 2018 - a oggi

- Automating OSINT
- Digital Healthcare
- Information Disorder
- Industry 4.0

Research
Interests

RIATLAS
HEALTHCARE



Description/Goals



- Understanding the big data problems and opportunities
- Mastering the Big Data landscape
 - From architectural point of view
 - From technologies point of view
- Mastering the most important technologies raised under the umbrella of Big Data
 - For **storing** the data
 - For **processing**
- Enjoying yourself with team working by focusing on the development of Big Data enabled architecture for addressing big data processing needs by using studied technologies.

Description/Goals - TOPICS



- Storage
 - Big Data Storage
 - Big Data File Formats
 - NoSQL Database
- Batch Processing: MapReduce, Spark
- Stream Processing: Spark Structured Streaming & Kafka Streaming

This course focuses on requirements, computational paradigms, systems, and architectures for Big Data,
not on data analysis techniques.

Description/Goals - TECHNOLOGIES



About the lessons plan



- Most of the lessons are organized including a practical session, but it depends on the topic and time
 - 2h and 15'
 - ❖ 2/3 (more or less) Teaching
 - ❖ 1/3 (hopefully) Hands-on explained technology
- Lessons during which each team will introduce the project results
 - I could give you Major or Minor Revision.
 - On the date of the exam you will show the final project.
- During the course I will ask you Reflections and Homeworks.



About the lessons



- 1st Lesson
 - Introduction
 - Lambda Architectures and Big Data Requirements
- 2nd Lesson (Storage)
 - Hadoop & HDFS
 - Big Data File Formats
- 3rd Lesson:
 - Complementary tools for collecting data
- 4th, 5th, 6th Lessons (**Batch Processing**)
 - Pig, Hive, Apache Spark and Pyspark
- 7th, 8th Lessons (**Schemless Storage**)
 - NoSQL DB
- 9th, 10th, 11th Lesson (**Stream Processing**)
 - Stream Processing: Spark Streaming, Apache Storm, Apache Kafka
- 12th, 13th Lessons:
 - Complementary tools for showing data
- 14th (Last Lesson): Project work presentations



About the lessons plan (2)



+4 Lessons:

- 2 - Vector DB
- 2 - LLM & RAG

+3 Lessons:

- Teamworking



About Exam



Team Work &
Discussion



Oral Discussion
Starting from Team Work



About Exam – Team Working



- Team composed of at least **2** persons and at most **4**
- Project Goals
 - Data Ingestion (made by yourself with our help)
 - **Data Processing**
 - **Data Storing/Indexing**
 - Data Visualization



About Exam – Team Working



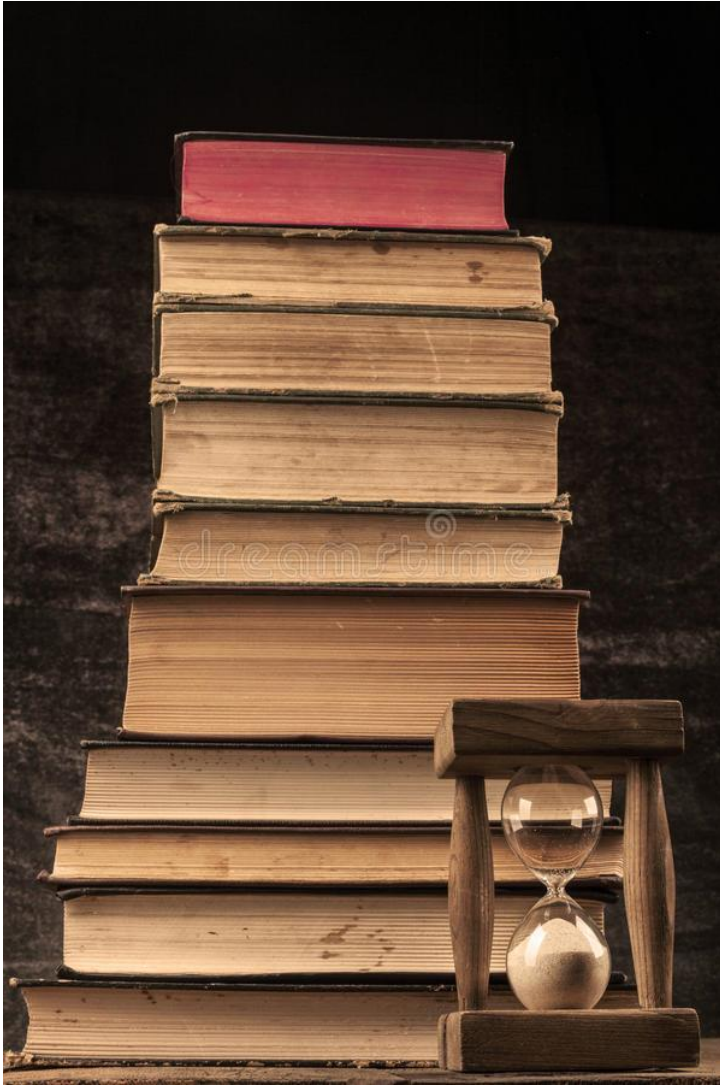
- Application domains:
 - Wearable Sensors,
 - OSINT,
 - IoT
- Accordingly,
 - some ideas at the end of this lesson,
 - your ideas for the next lesson
- I will show you some examples



About Exam



- Deadlines are important:
 - You can reach 30 e lode only until Feb 2025
 - Minus 2 points at each exam session



Some Facilities



- In the previous years, I prepared a Virtual Machine for each technology we will use during the course, you can reach it here
 - <https://drive.google.com/drive/folders/113fCoZK2tYeWvyvkoUKRpYdQEtslbsOZ>
- This year, I strongly suggest to become familiar with **docker** images, I will give few instructions at the end of lesson
- I will share slides of each lesson, possibly before each lesson



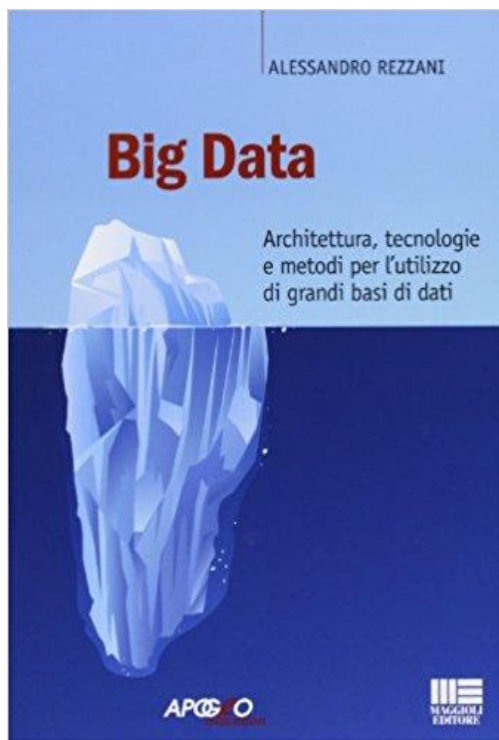
Some Facilities



- Cloud & Containerization:
 - **Docker**
 - Kubernetes



BigData... choosing the right books is not easy!

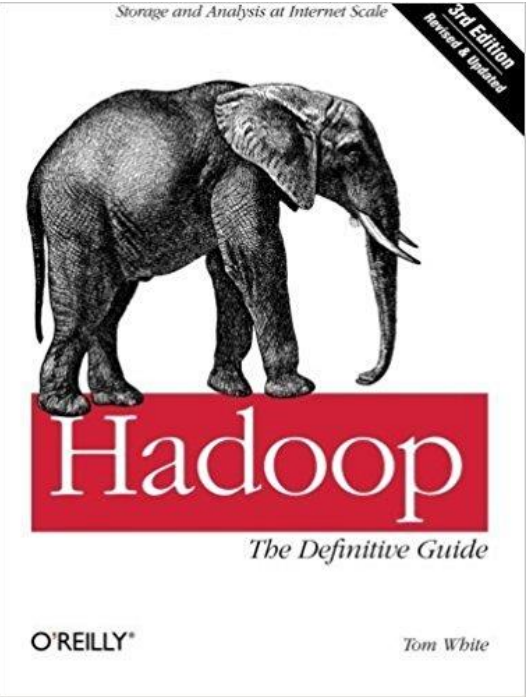
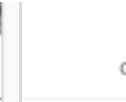


Big Data
Architettura, tecnologie e metodi per
l'utilizzo di grandi basi di dati

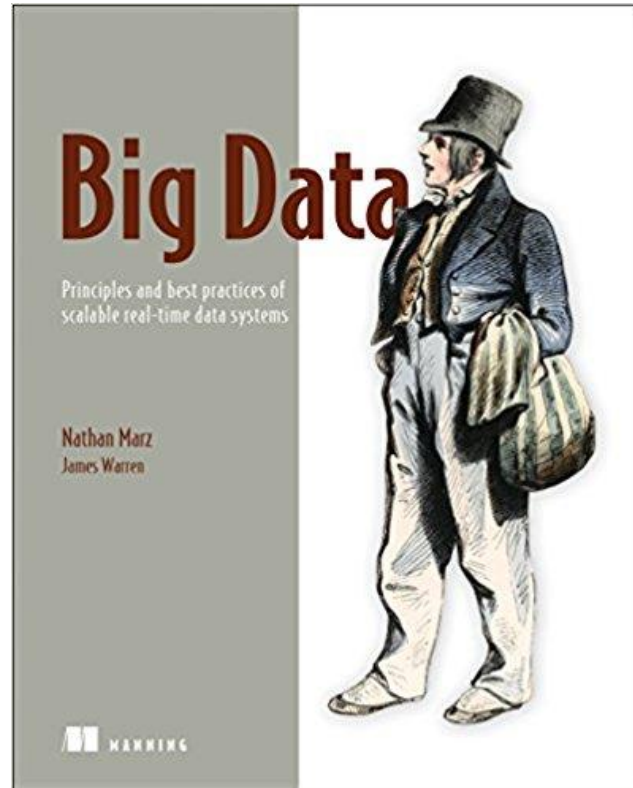
- *Alessandro Rezzani*
- *Apogeo Editore*

BigData... choosing the right books is not easy!



							
Big Data: A Revolution That Will Transform How We Live, Work, and Think 2013	Hadoop: The Definitive Guide, 4th Edition Storage and Analysis at Internet Scale By <u>Tom White</u> Publisher: <u>O'Reilly Media</u> Release Date: April 2015 Pages: 235	Spark: The Definitive Guide 2015	Big Data @ Work: Dispersing the Power of Analytics Thomas H. Davenport	The Human Face of Big Data Rick Smolan...	Too Big to Ignore: The Big Data Revolution Phil Simon, ...	Everybody Lies: Big Data and the End of Privacy Seth Stephens-Davidowitz	Doing Data Science 2013

BigData... choosing the right books is not easy!



Big Data

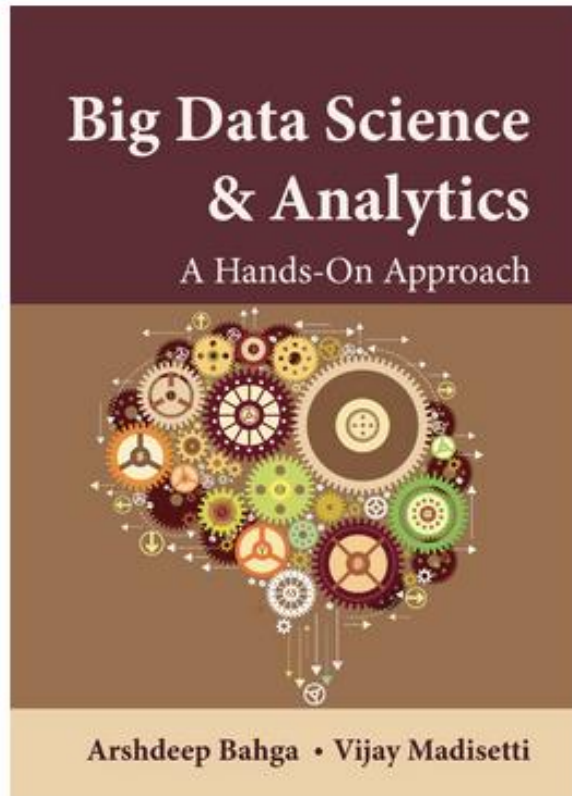
Principles and best practices of scalable realtime data systems

Nathan Marz and James Warren

April 2015 · ISBN 9781617290343 · 328 pages · printed in black & white

I love this book!

BigData... choosing the right books is not easy!

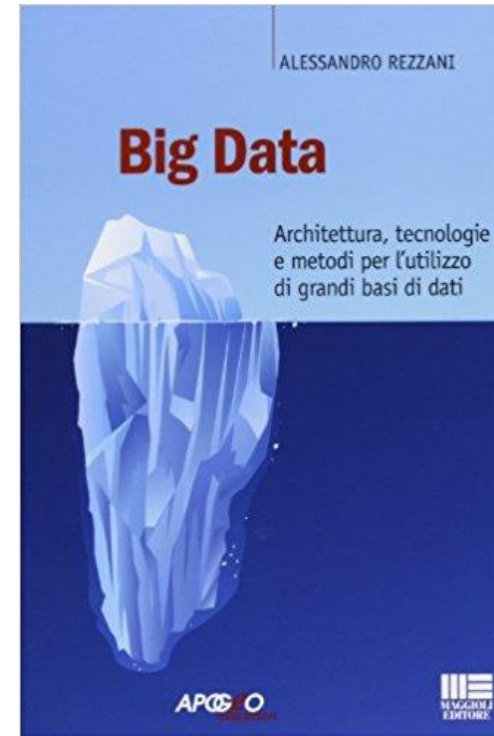
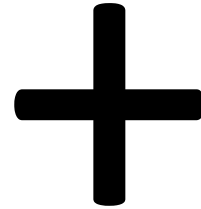
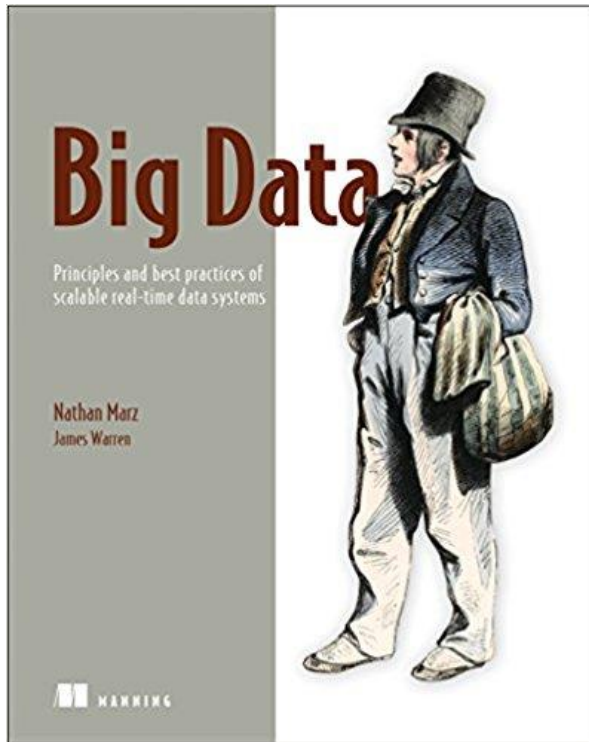


Chapter 1 is really useful for Big Data Introduction.

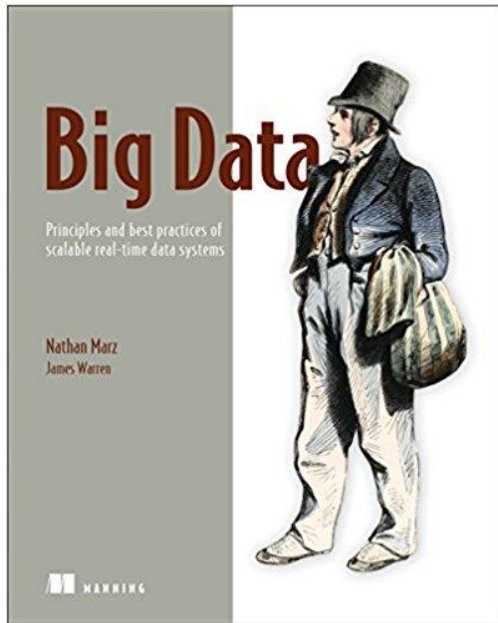
Hands-on Approach

<http://www.hands-on-books-series.com/bigdata.html>

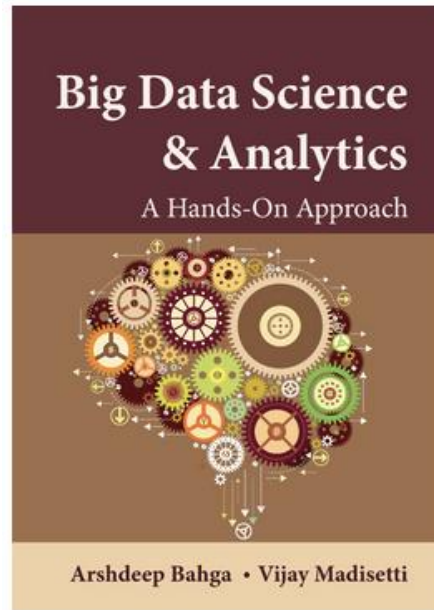
In my case...



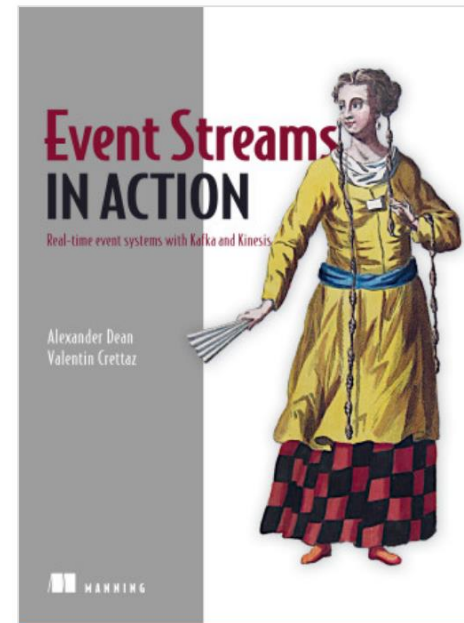
What I suggest you...



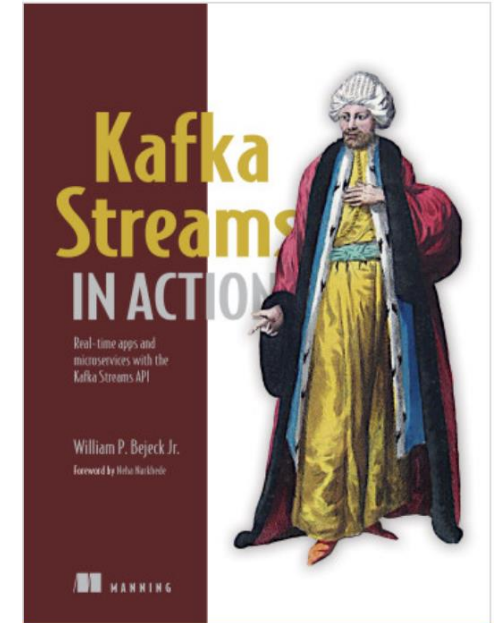
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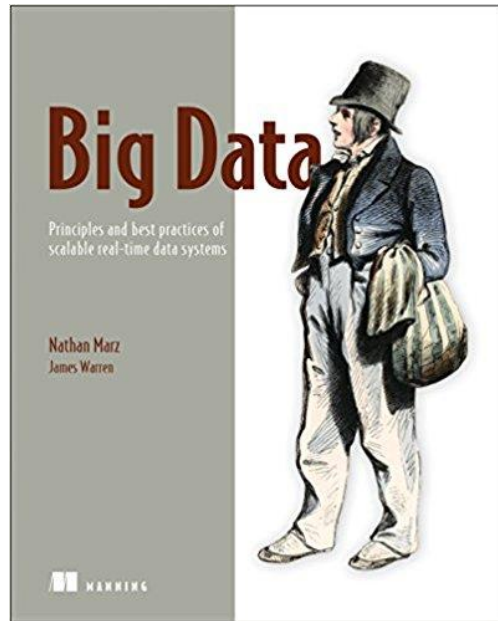
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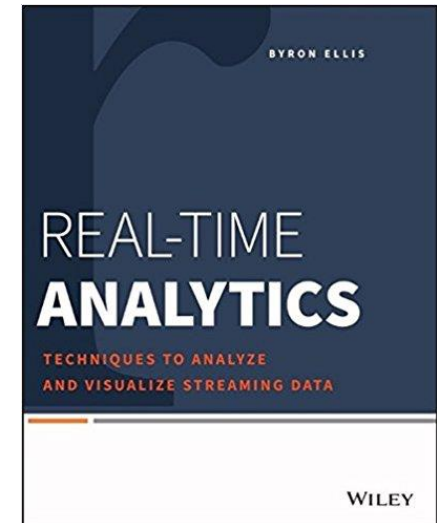
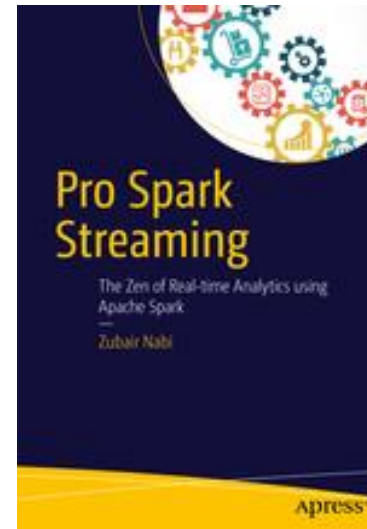
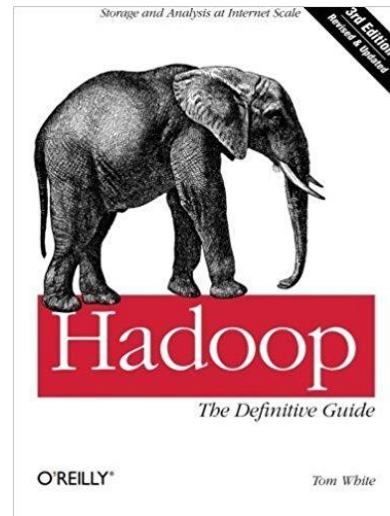
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Additionally, there are many manuals



+





Let's start...

1st Lesson – Roadmap



- About
 - me
 - lesson
 - exam
- Big Data
 - Introduction Big Data
 - Big Data Opportunity
 - Big Data Information System
 - Big Data-enabled Architecture: Lambda and Kappa



Quotes About Big Data



- Big
 - “The greater the struggle, the more glorious the triumph.” (Butterfly Circus)
- Necessary
 - “It is a capital mistake to theorize before one has data.” (S. Holmes)
- Fashionable
 - “I always wanted to be fashionable.” (J. Malkovich)
- **Profitable**
 - **“Data is a precious thing and will last longer than the systems themselves.” (T. Berners-Lee)**
- Exciting
 - “The most exciting phrase to hear in science, is not ‘Eureka!’, but ‘That's funny’...” (I. Asimov)

Quotes About Big Data



- “Hiding within those mounds of data is knowledge that could change the life of a patient, or change the world.” (Atul Butte, Stanford)
- “Information is the oil of the 21st century, and analytics is the combustion engine” (Peter Sondergaard, Senior Vice President, Gartner)
- “War is 90% information.” (Napoleon Bonaparte)
- “Big Data will spell the death of customer segmentation and force the marketer to understand each customer as an individual within 18 months or risk being left in the dust.” (Ginni Rometty, CEO, IBM)
- “Without big data analytics, companies are blind and deaf, wandering out onto the web like deer on a freeway.” (Geoffrey Moore, author and consultant)
- “The temptation to form premature theories upon insufficient data is the bane of our profession.” (Sherlock Holmes, fictional detective)
- "The most valuable commodity I know of is information." (Gordon Gekko, Wall Street)

- The term of *big data* was coined under the explosive increase of global data and was mainly used to describe these **enormous datasets**.
- **Traditional database systems**, such as relational databases, have been pushed to the limit. In an increasing number of cases these **systems are breaking under the pressures of "Big Data."**
- **Traditional systems**, and the data management techniques associated with them, **have failed to scale to Big Data**.
- **A new breed of technologies has emerged**. Many of these new technologies have been grouped under the term "NoSQL." In some ways these new technologies are more complex than traditional databases, and in other ways they are simpler. These systems can scale to vastly larger sets of data, but using these technologies effectively requires a fundamentally new set of techniques. **They are not one-size-fits-all solutions**.
- **Many of these Big Data systems were pioneered by Google**, including distributed filesystems, the **MapReduce** computation framework, and distributed locking services. Another notable pioneer in the space was **Amazon**, which created an innovative distributed key-value store called **Dynamo**. The **open source community** responded in the years following with **Hadoop**, **HBase**, **MongoDB**, **Cassandra**, **RabbitMQ**, and **countless other projects**.

Infrastructure

Analytics

Applications

Hadoop On-Premise: cloudera, Hortonworks, MAPR, Pivotal, IBM InfoSphere, bluedata, jethro

Hadoop in the Cloud: amazon web services, Microsoft Azure, Google Cloud Platform, IBM InfoSphere, CAZENA, TREASURE DATA, altiscale, Qubole

Spark: databricks, GridGain, TACHYON NEXUS

Cluster Services: amazon web services, kubernetes, HPCC SYSTEMS, docker, MESOSPHERE, Core OS, pepperdata, StackIQ

Analyst Platforms: Palantir, AYASDI, Quid enigma, Digital Reasoning, ORBITAL INSIGHT

Analytics Platforms: Microsoft, guavus, Datameer, Bottlenose, inter|ana

Data Science Platforms: context relevant, CONTINUUM ANALYTICS, DataRobot, Alpine, MODE, plotly, ARIMO, dataiku, nu|tonian, DOMINO, yhat, sense, ALGORITHMIA

Visualization: +tableau, Google Cloud Platform, Qlik, looker, Roambi, SISENSE, VIZOMDATA, datorama, CHARTIO

Sales & Marketing: RADIUS Gainsight, bloomreach, Zeta, EVERSTRING, blueyonder, livefyre, Lattice, kahuna, infer, SAILTHRU, persado, AVISO, usense, QUANTIFIND, ACTIONIQ, fuse|machines, JE NGAGIO

Customer Service: MEDALLIA, ATTENSTITY, CLARABRIDGE, CLICKFOX, STELLAService, NGDATA, Preact, DigitalGenius, apurva|usep

Human Capital: gild, Connectix, eBrevia, PREMONITION

Legal: RAVEL, MEDICATA, Everlaw, Bredia

NoSQL Databases: amazon DynamoDB, Google Cloud Platform, Microsoft Azure, ORACLE, mongoDB, MarkLogic, DATASTAX, Couchbase, SequoiaDB, redislabs, influxdata

NewSQL Databases: SAP HANA, Clustrix, Pivotal, paradigm4, nuODB, memsql, splice MACHINE, MariaDB, VOLTDB, citusdata, deapdb, Trafodion, Cockroach LABS

BI Platforms: Power BI, amazon web services, Domo, Wave Analytics, GoodData, birst, kyvos insights, platforma, atscale, ARCADIA, SISENSE

Statistical Computing: sas, SPSS, MATLAB

Log Analytics: splunk, sumologic, kibana, CLOUD PHYSICS, loggly

Social Analytics: Hootsuite, NETBASE, DATASIFT, tracx, bitly, synthesio, simple reach

Ad Optimization: AppNexus, MediaMath, critico, OpenX, Integral Ad, TradeDesk, Levelintest, DataXu, Uppler, MOAT

Security: CounterTrack, cyberason, ThreatMetrix, SentinelOne, Recorded Future, Guardian Analytics, FORTSCALE, sift science, Keybase, feedzai, SIGNIFYD

Vertical AI Applications: facebook, Clara, KASIST@, lumiata

Graph Databases: neo4j, GIRAAPH, OrientDB, InfiniteGraph

MPP Databases: TERADATA, VERTICA, An HP Company, NETEZZA, Action, koginitio, EXASOL, dremio

Cloud EDW: amazon web services, Google Cloud Platform, Microsoft Azure, Pivotal, snowflake, WATERLINE DATA, Infoworks

Data Transformation: alteryx, talend, TRIFACTA, tamr, Paxata, StreamSets, Alation

Data Integration: informatica Put potential to work:, MuleSoft, snapLogic, Bedrock Data, xplenty

Real-Time: amazon web services, METAMARKETS, strim, confluent, DATAPREPINT, dataArtisans

Machine Learning: Azure Machine Learning, amazon web services, P2Ogi, SKY TREE, rapidminer, DATARM, deepsense, VISEN, Prediction, IDIBON, yseop

Speech & NLP: NarrativeScience, NUANCE, semantic, semantix, nora, HyperScience, SI, LEADED INTERFACE, clarifai, DEXTRO, MetaMind

Horizontal AI: Warpp, sentient technologies, vicarious, nora, HyperScience, SI, LEADED INTERFACE, clarifai, DEXTRO, MetaMind

Publisher Tools: Outbrain, Taboola, quantcast, Chartbeat, yieldbot, Yieldmo

Govt / Regulation: Socrata, OPENGOV, FN FiscalNote, enigma, mark43, PREDPOL, OpenDataSoft

Finance: Affirm, LendingClub, OnDeck, Kreditech, zest finance, LendUp, Kabbage, Insikt, tidemark, Payoff, uora, Dataminr, Lendio, KENSHC, AIDYA, iSENTIUM, Quantopian

Education/ Learning: KNEWTON, Clever, Declara, PANORAMA, knowtre

Life Sciences: 23andMe, Counsyl, PATHWAY GENOMICS, Recombine, deep genomics, KYRUUS, FLATIRON, zymergen, HealthTap, METABIOTA, ZEPHYR HEALTH, ovla, Ginger.io, transcriptic, Glow, enlitic, AiCure, Atomwise

Industries: OPOWER, eHarmony, RetailNext, duetto, STITCH FIX, WorkFusion, BLUE RIVER, TACHYUS, Seeq, FarmLogs, SwiftKey, relect, SIGHT MACHINE, statmuse, BOXEVER

Cross-Infrastructure/Analytics: amazon web services, Google, Microsoft, SAP, sas, data, hp, Autonomy, VERTICA, VMware, TIBCO, TERADATA, ORACLE, NetApp

Open Source

Open Source

The banner displays a wide array of tools and frameworks used in data science and big data, organized into nine categories:

- Frameworks:** Hadoop, HDFS, YARN, MapReduce, Tez, Flink, CDAP.
- Query / Data Flow:** SLAM DATA, Apache Drill, Google Cloud Dataflow.
- Data Access:** Cassandra, HBase, Accumulo, MongoDB, CouchDB, Riak, QSciDB, Kafka, Nifi, Opentsdb.
- Coordination:** Apache Zookeeper, Apache Ambari.
- Real-Time:** Storm, Spark, APEX, Flink, Tachyon, Druid.
- Stat Tools:** R, Scalalab, NumPy, SciPy.
- Machine Learning:** Mllib, Aerosolve, Caffe, Weka, FeatureFu, DIMSUM, Jupyter, DL4J, Mahout, CNTK, TensorFlow.
- Search:** Elasticsearch, Solr, Lucene.
- Security:** Apache Ranger, Zeppelin.

A large red text "A plethora" is written diagonally across the banner.

Data Sources & APIs

The banner displays logos for various companies across six categories:

- Health:** Apple, Jawbone, Garmin, Practice Fusion, Fitbit, Withings, Validic, Netatmo, Kinsa, Human API.
- IOT:** Uptake, ThingWorx, Helium, Samsara, August, Estimate.
- Financial & Economic Data:** Bloomberg, Thomson Reuters, Dow Jones, Yodlee, Premise, S&P Capital IQ, Quandl, Xignite, CB Insights, Mattermark, StockTwits, Estimize, Plaid.
- Air / Space / Sea:** Planet Labs, Spire, Windward, Cruise, SkyCatcher, Airware, DroneDeploy.
- Location / People / Entities:** Acxiom, Experian, Epsilon, InsideView, Esri, Foursquare, Streetline, Crimson Hexagon, CartoDB, Factual, PlaceIQ, Circulate, Placemeter, Basis Technology, Sense360.
- Other:** Qualtrics, Panjiva, Data.gov, GA, Pluralsight, DataCamp, Insight, DataElite, The Data Incubator, Metis.



Big Data and Big Data Analytics

What is big data?



Big Data is
any thing
which is
crash Excel.

Small Data is
when is fit in RAM.
Big Data is when is
crash because is
not fit in RAM.



Or, in other words, Big Data is data in volumes too great to process by traditional methods.

- “Big data refers to data sets whose size is **beyond** the ability of typical database software tools to capture, store, manage and analyze.” *The McKinsey Global Institute, 2012*
- “Big data is mostly about taking numbers and using those numbers to **make predictions about the future**. The bigger the data set you have, the more accurate the predictions about the future will be.” *Anthony Goldbloom, Kaggle’s founder*

Originally 3Vs

VOLUME

- Terabytes
- Records
- Transactions
- Tables, files

3 Vs of Big Data

- Batch
- Near time
- Real time
- Streams

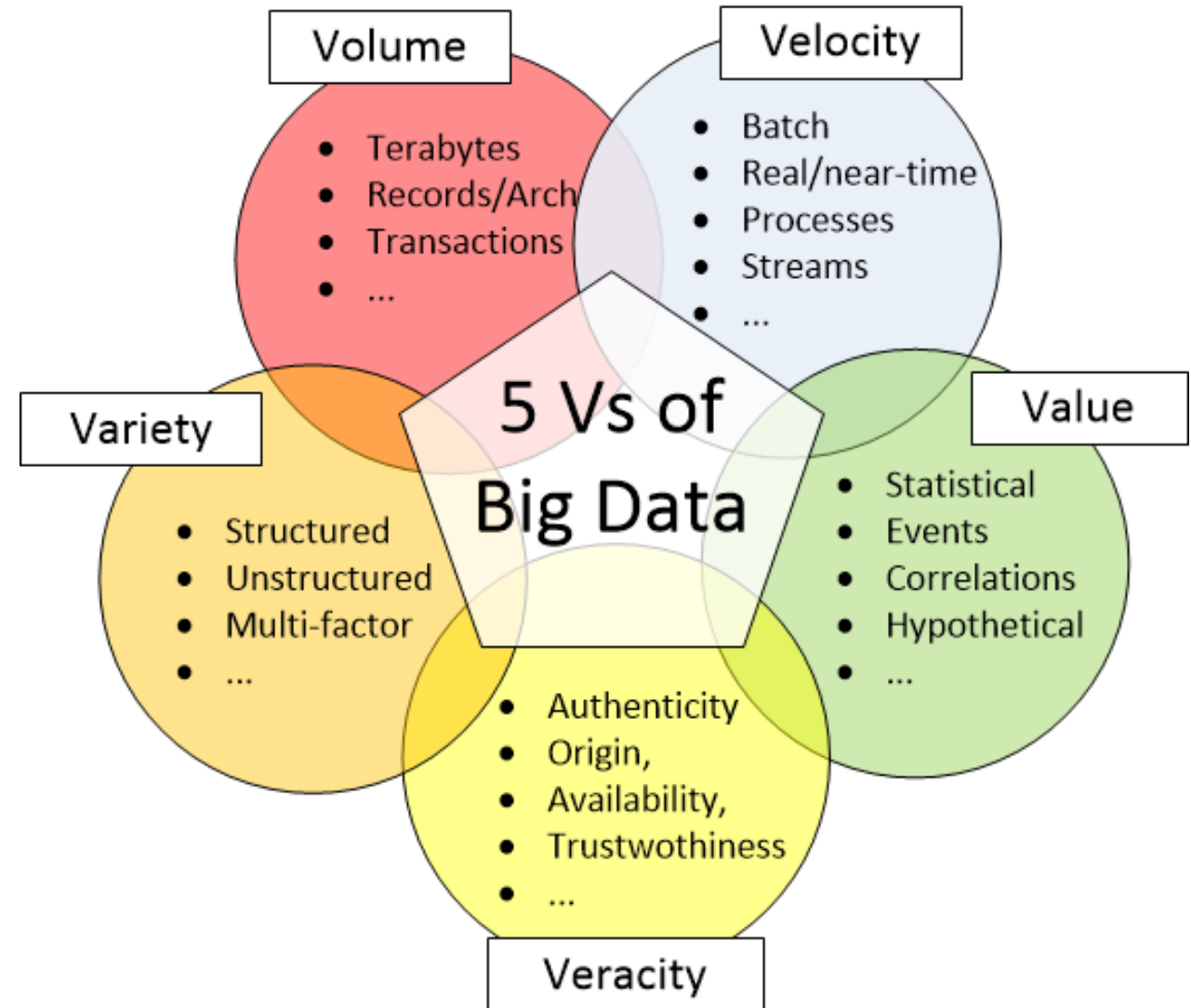
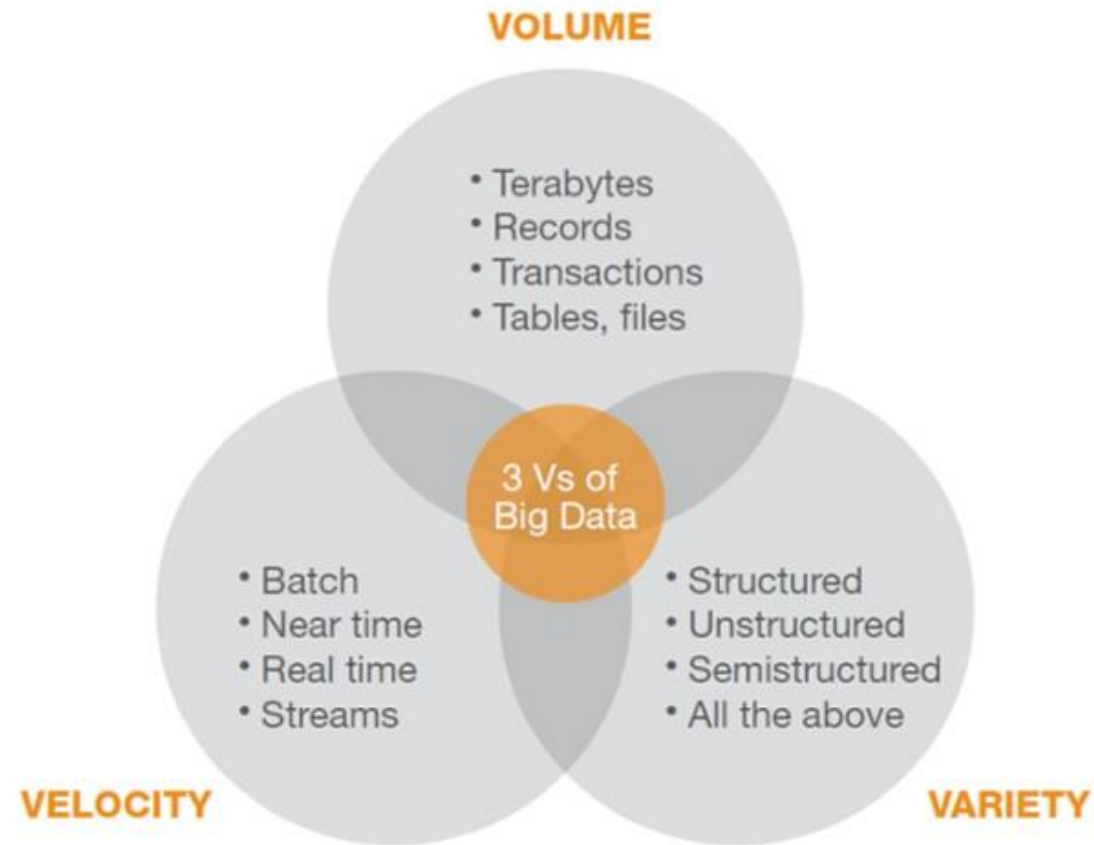
- Structured
- Unstructured
- Semistructured
- All the above

VELOCITY

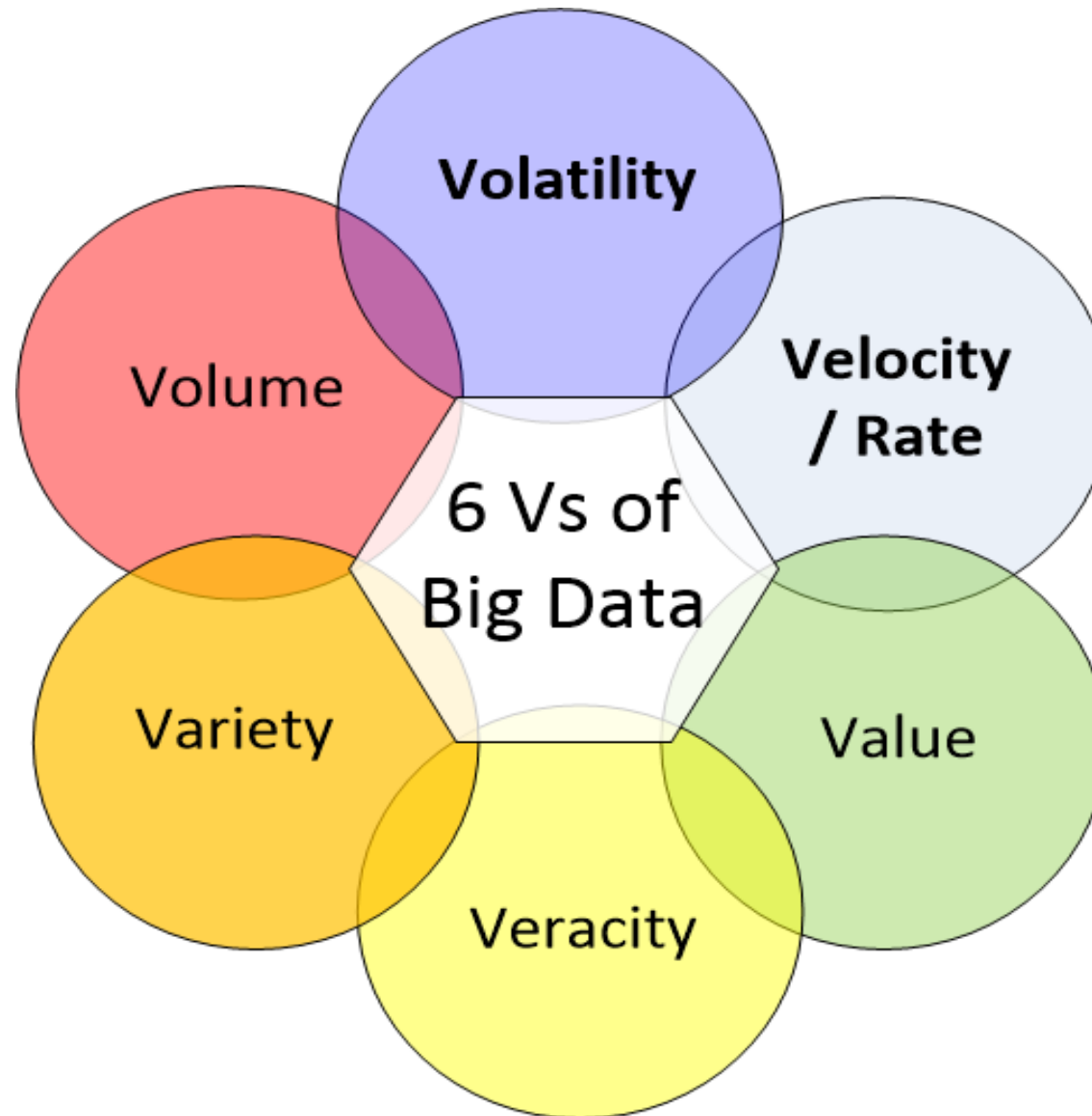
VARIETY



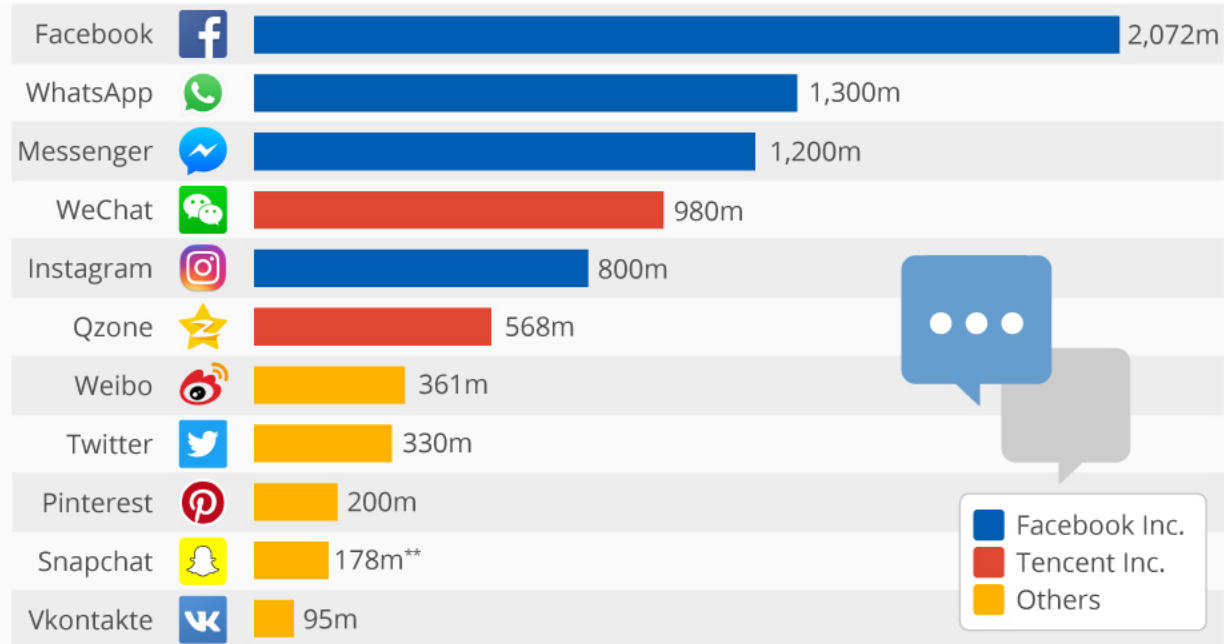
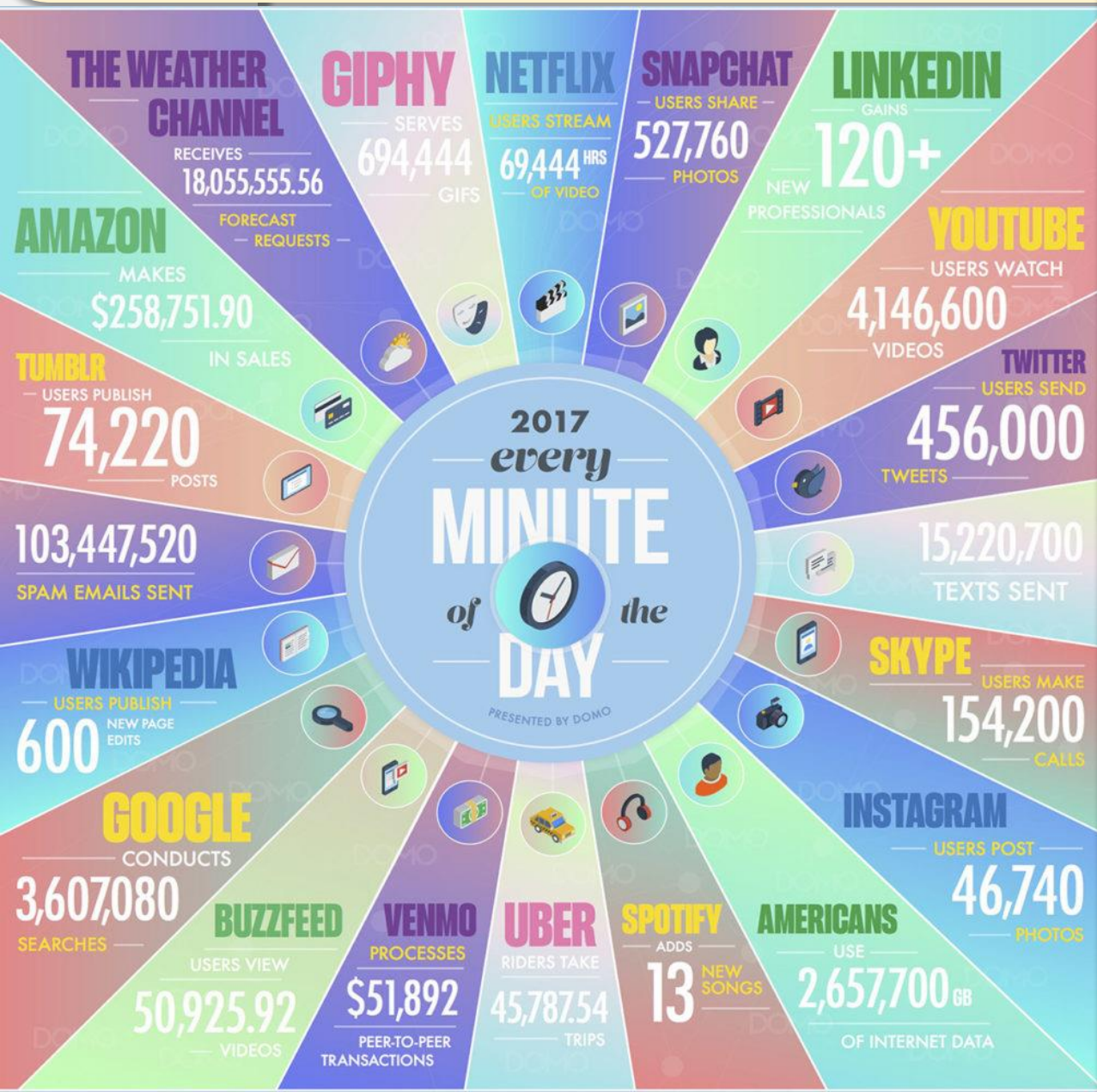
The 3Vs...5Vs...6Vs of Big Data



One more V...



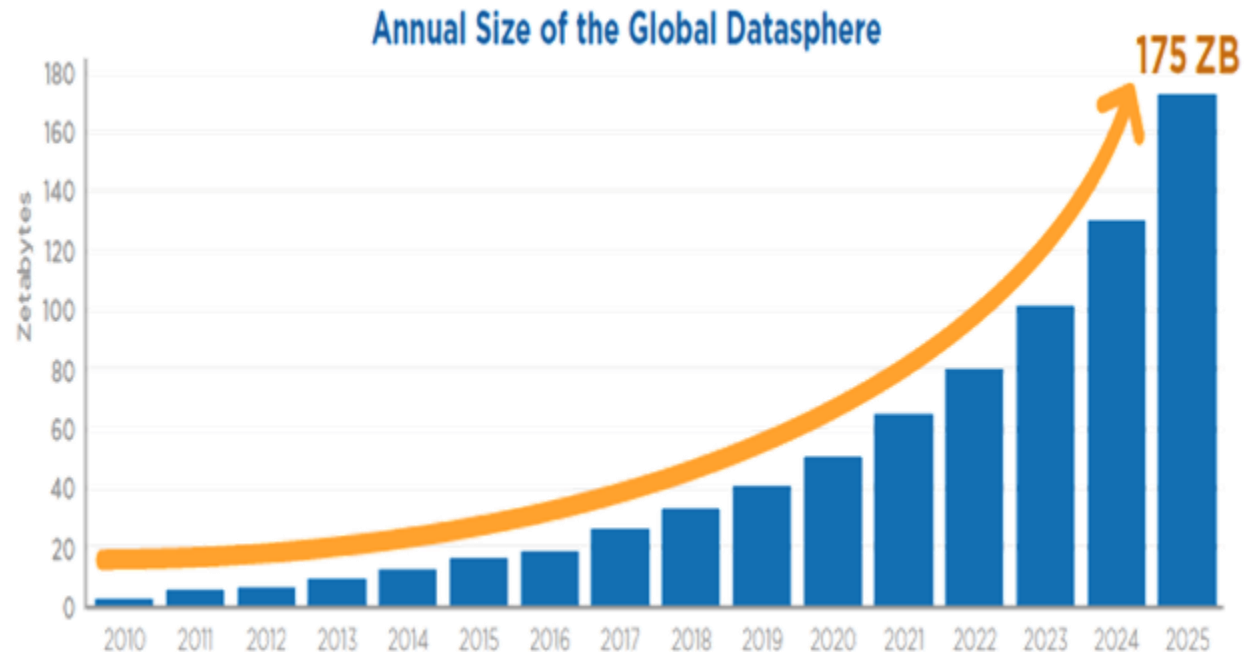
Why now?



How much data?

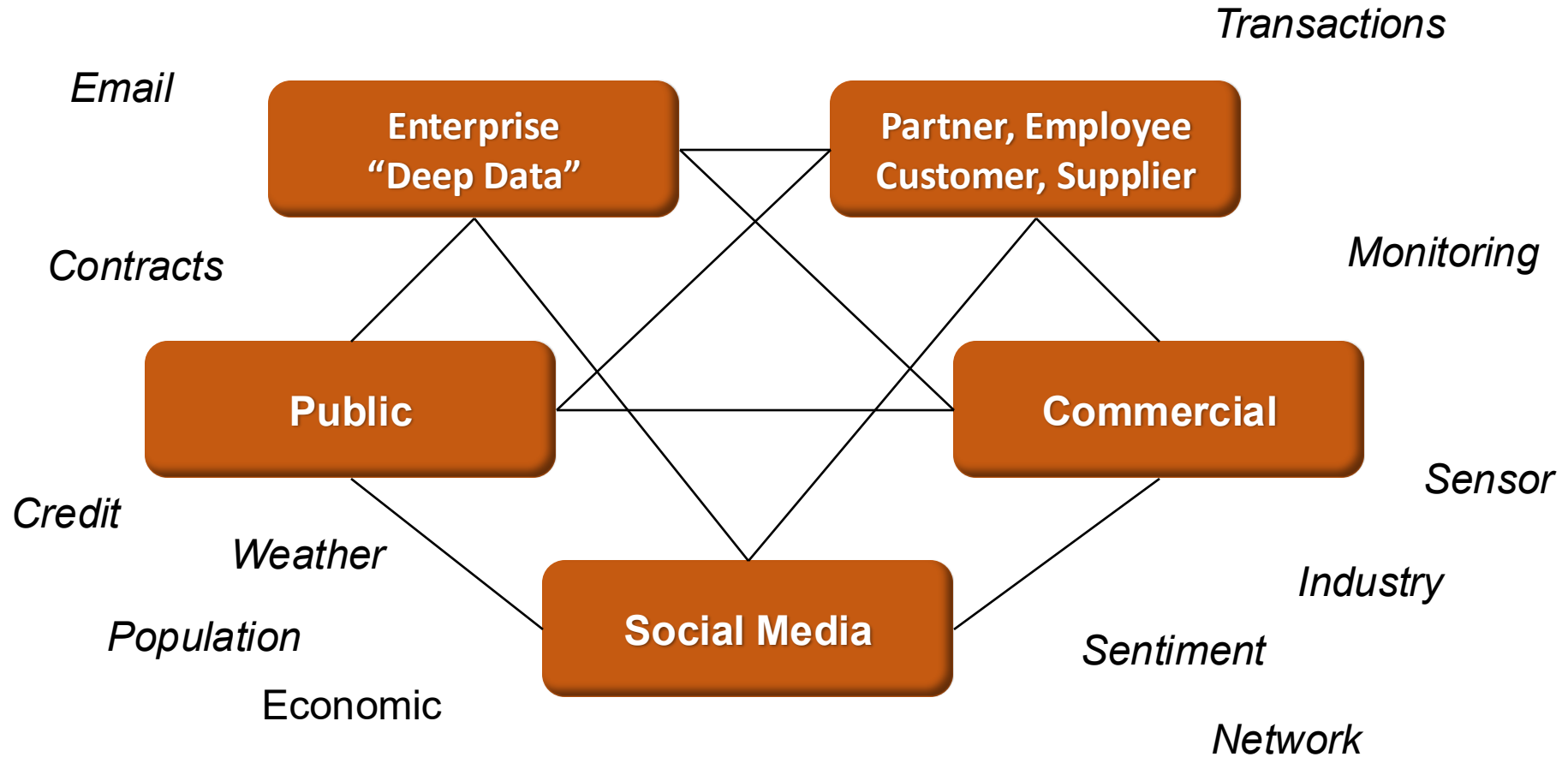


- Datavolume: from Terabytes to Zettabytes
 - How big is a Zettabyte? <http://bit.ly/2G15uVI>
- 40 Zettabytes of data will be around in 2020
 - 40 Zettabytes ($40 \times 10^{21} \approx 40 \times 270$ Zettabytes) ...
 - 40000 Exabytes (40000×10^{18}) ...
 - 40000000 Petabytes (40000000×10^{15}) ...
 - 40000000000 Terabytes ($40000000000 \times 10^{12}$) ...
 - 400000000000000000000000 bytes!
- Data will be increasingly stored in the public cloud.



Source: Data Age 2025, sponsored by Seagate with data from IDC Global DataSphere, Nov 2018

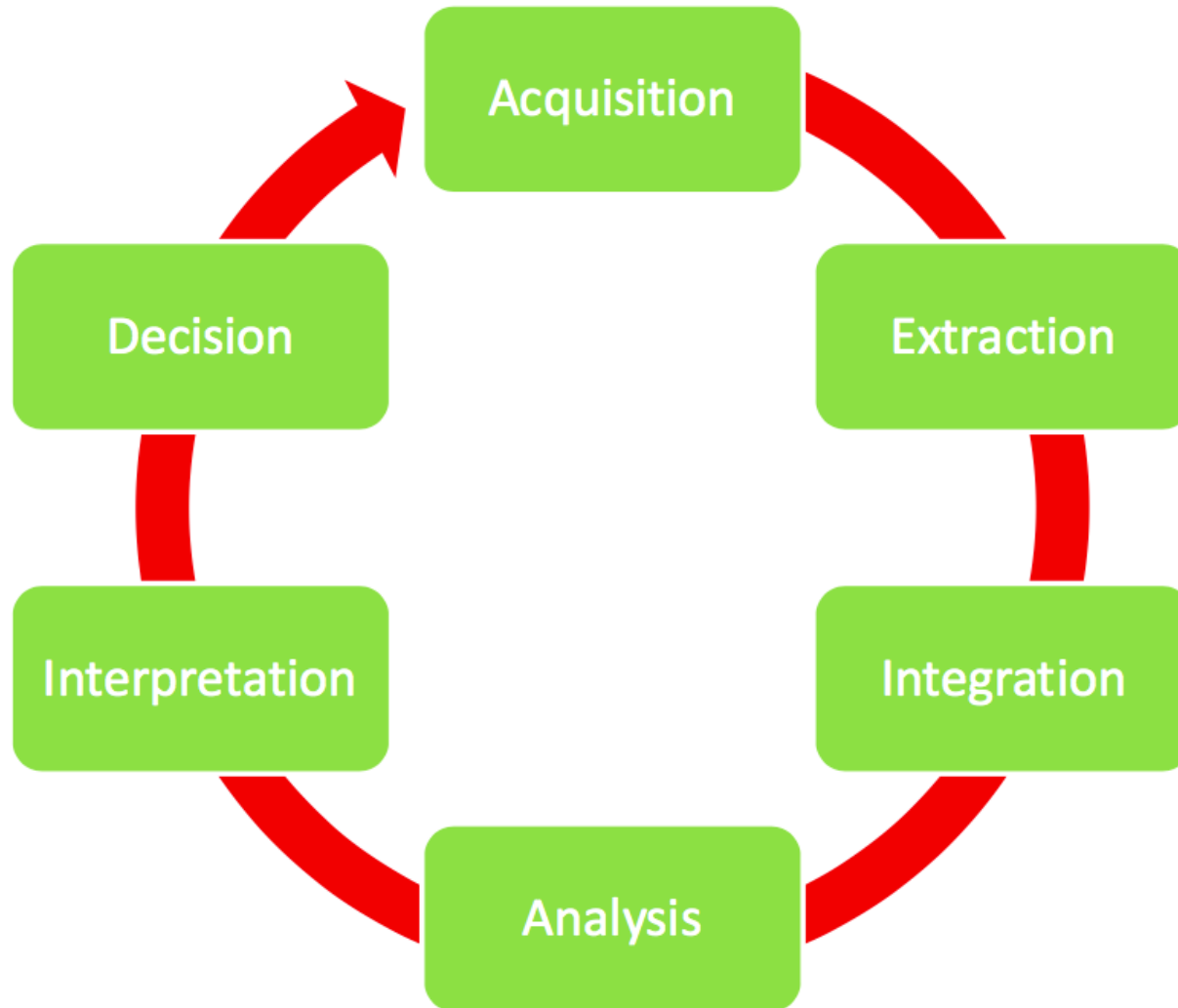
Where are Big Data?



Data Analytics - Opportunities



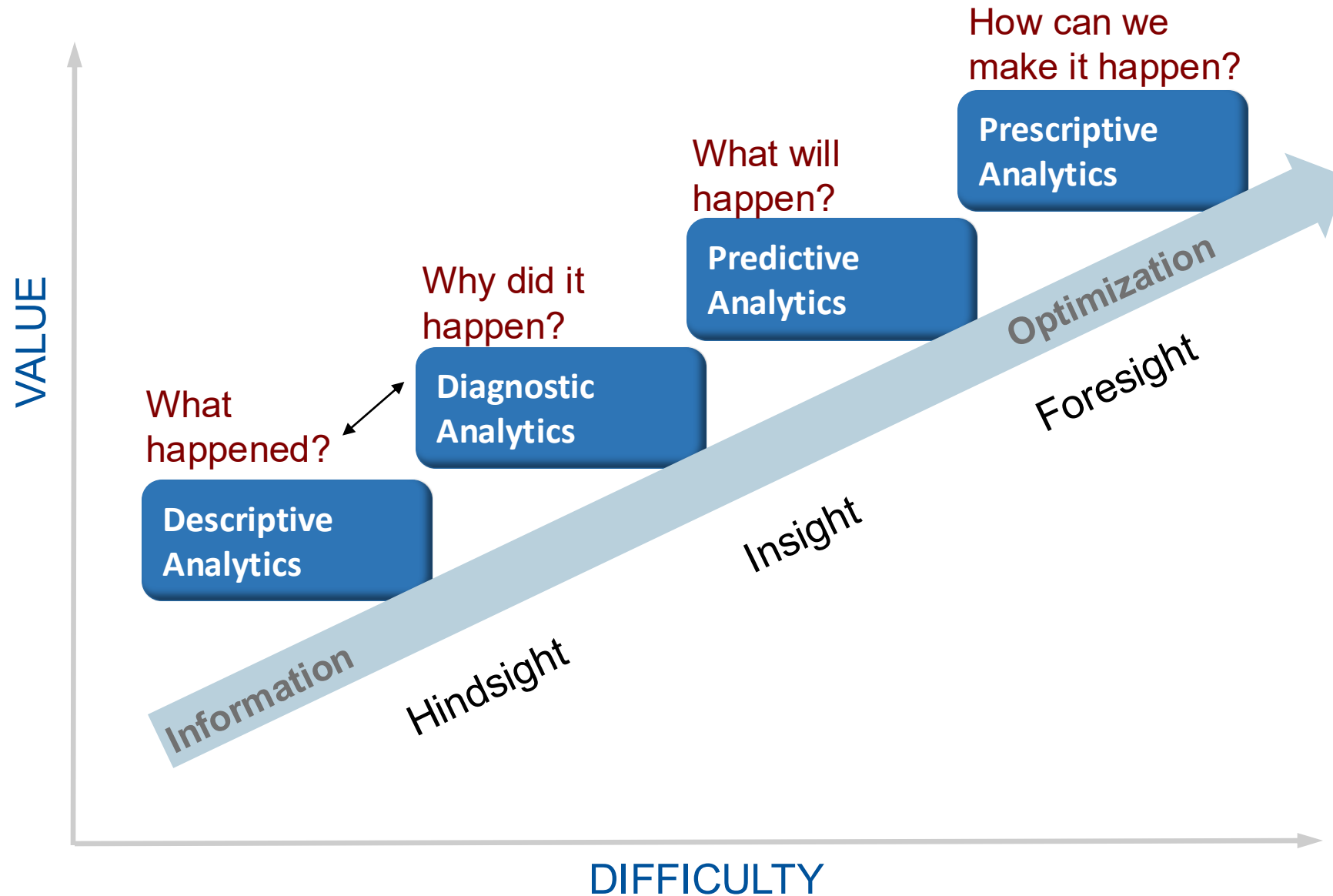
- In September 2004, by analyzing 460 terabytes of data in their data warehouse, when Hurricane Charley hit the Florida Gulf Coast, Walmart was able to predict what products people in Miami would want to buy.
- By studying a market basket of what individuals buy, Wal-Mart can set prices to attract customers who want to buy “loss leader” items because they will also likely put several higher-margin products in the same shopping cart.
- Predict how many cashiers are needed at different hours in different stores given the time of year, holidays, weather, pricing, and many other factors.
- ... and much more.



- ***Decision Making Support***

- *knowledge of the domain*
- *knowledge of the provenance*
- *identification of patterns of interest*
- *flexibility of the process*

Big Data Analytics - Models



Big Data Analytics - Models



Techniques

- Suggested Readings: Proactive Analytics
 - <https://www.christopherspenn.com/2018/02/the-next-evolution-of-analytics-is-proactive/>
- Looking for Big Data Failures
 - Google Flu:
 - ❖ <https://www.forbes.com/sites/stevensalzberg/2014/03/23/why-google-flu-is-a-failure/#5fbd62095535>
 - Search other Big Failures
 - ❖ ...

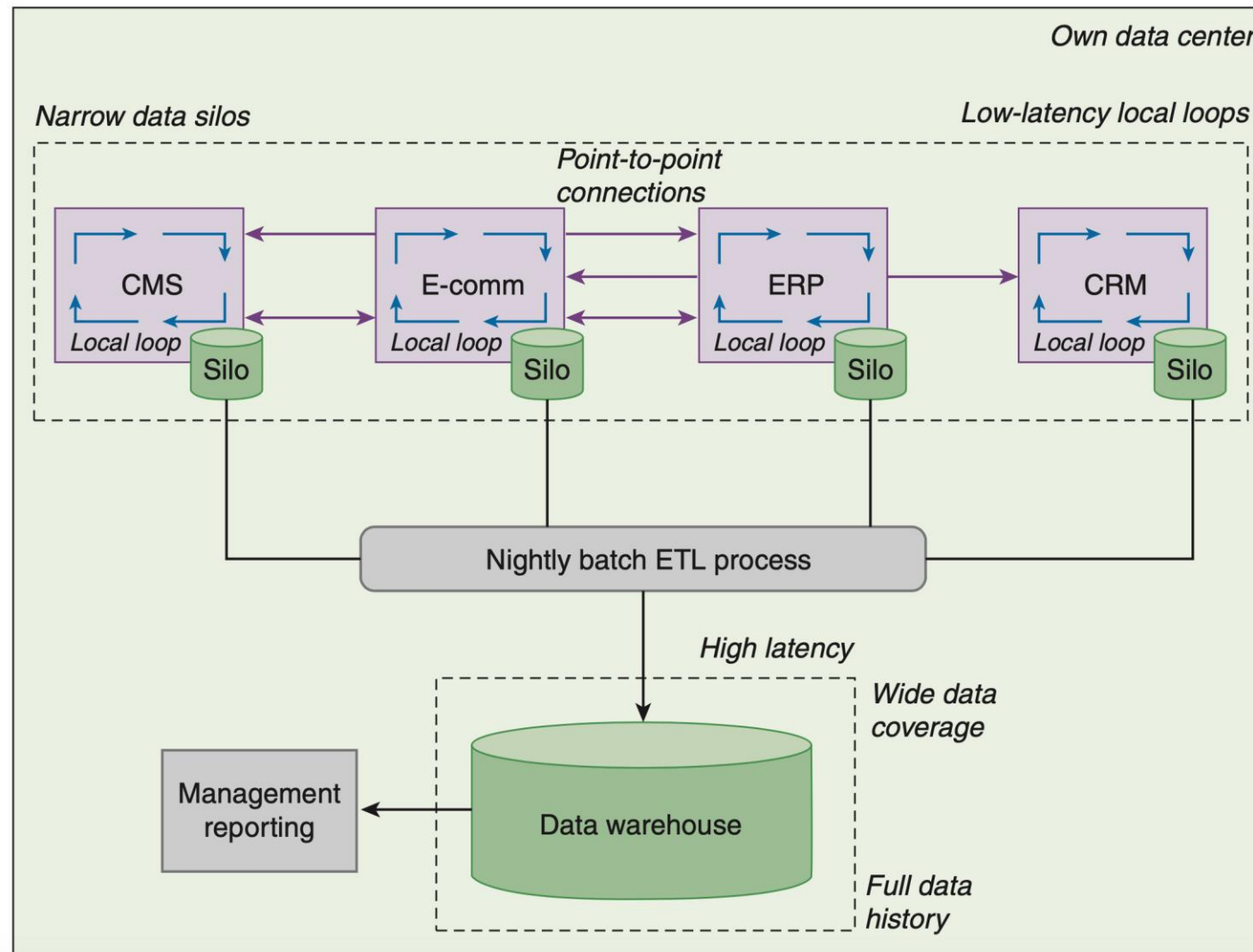
- Operational Processing → Informational Gap
- One-thing-at-a-time → Information Silos
- Decision makers need
 - Integrated View of Several Data Sources
 - Time-variant Data
 - Subject-oriented Analysis
 - ...

Operational vs Informational Processing



	Operational Systems	Informational Systems
Primary purpose	Run the business on a current basis	Support managerial decision making
Type of data	Current representation of state of the business	Historical point-in-time (snapshots) and predictions
Primary users	Clerks, salespersons, administrators	Managers, business analysts, customers
Scope of usage	Narrow, planned, and simple updates and queries	Broad, ad hoc, complex queries and analysis
Design goal	Performance: throughput, availability	Ease of flexible access and use
Volume	Many constant updates and queries on one or a few table rows	Periodic batch updates and queries requiring many or all rows

Typical DW Architecture



Batch vs Stream Processing

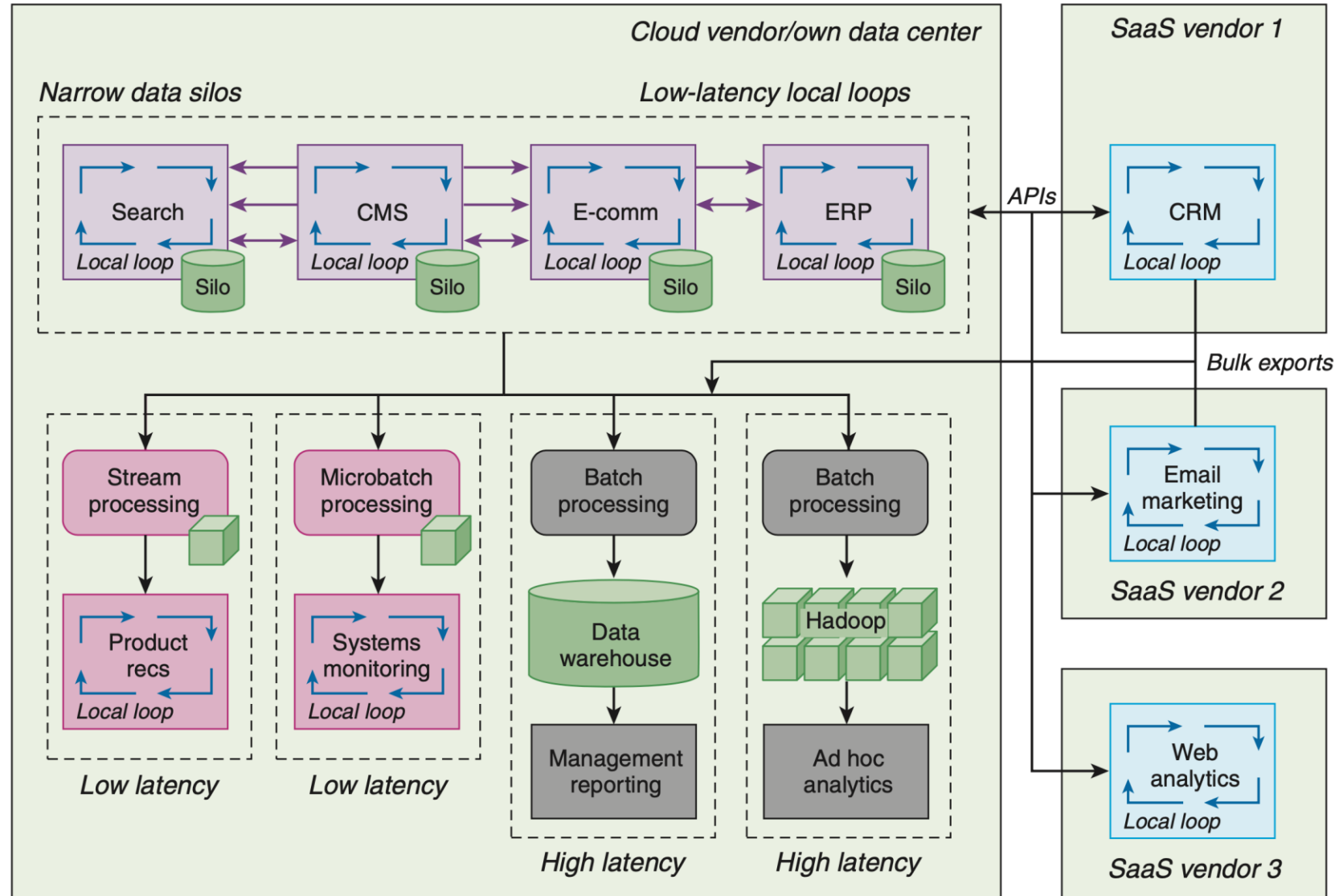


	Batch Processing (Data Lake)	Real-time Processing (Stream)
Log Analysis	What went wrong an hour ago?	Take corrective action / notify admin
Billing Analysis	How much did you spend during the last billing cycle?	Notify user that billing limit is approaching
Customer Preferences	Which ads were most effective yesterday?	Which ad should the customer see right now?
Fraud	Audit trail / forensic evidence	Stop fraud now
Smart Cities	Is it better to plan a new bus line? How to change the urban plan to reduce the traffic?	How to face down the occurring traffic jam near to the airport? How to allocate and coordinate suitable resources to manage the occurring emergency in the city center?

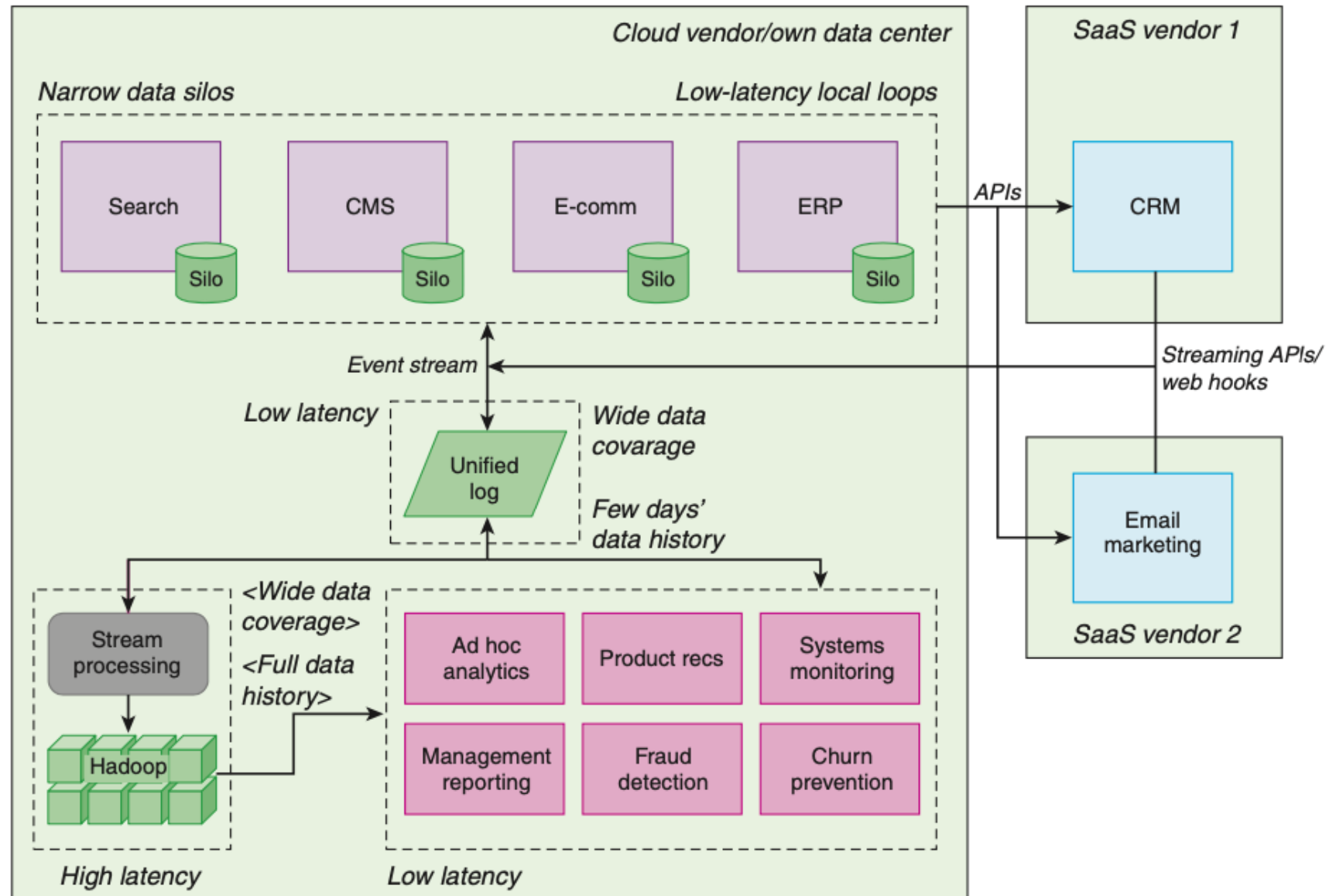
PLANNING

REACTING

Typical Emerging Architecture for Big Data



Next Trend: Architecture centered on unified log



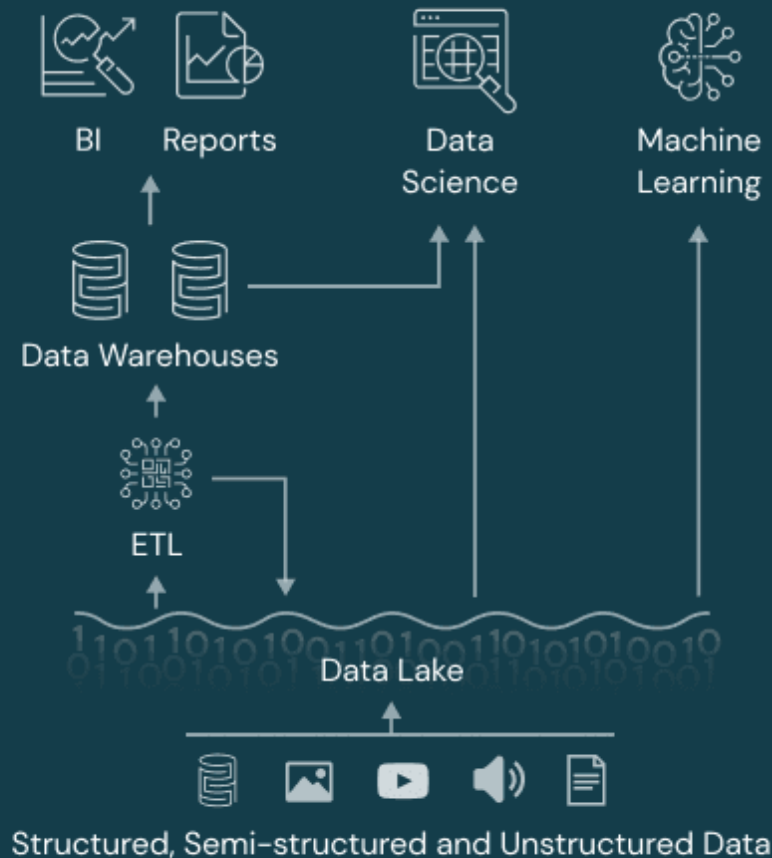
Cenni a Data Lake e Data Lakehouse



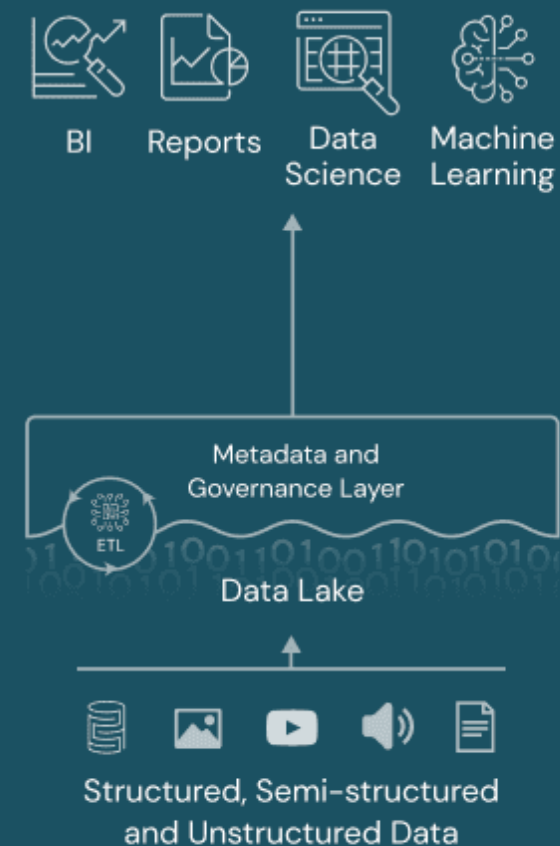
Data Warehouse



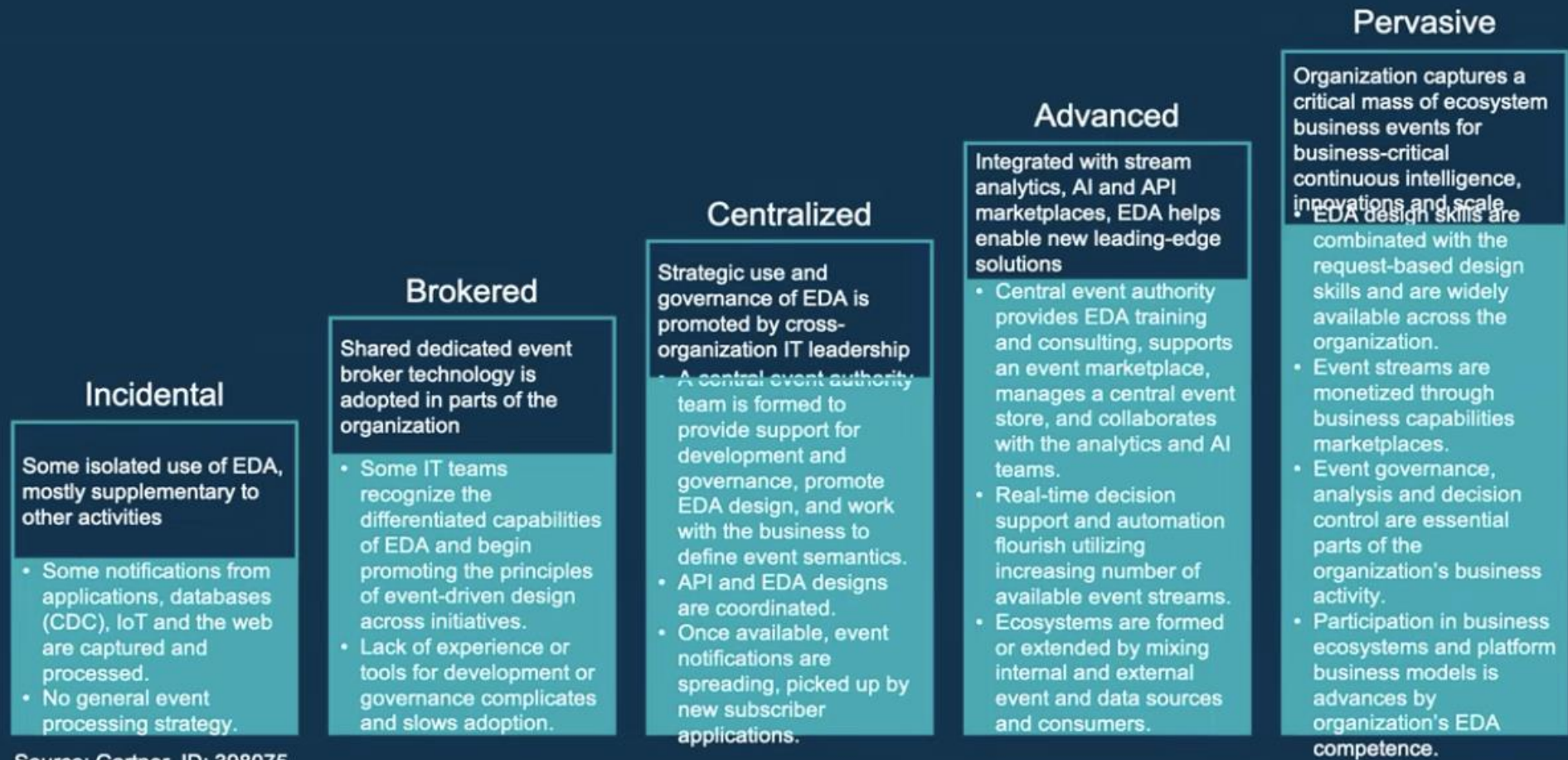
Data Lake



Data Lakehouse



Kafka and Event Driven Architecture



Source: Gartner, ID: 398075

Before Kafka

Local Kafka

Central Kafka

Kafka everywhere

Event-Driven Architecture

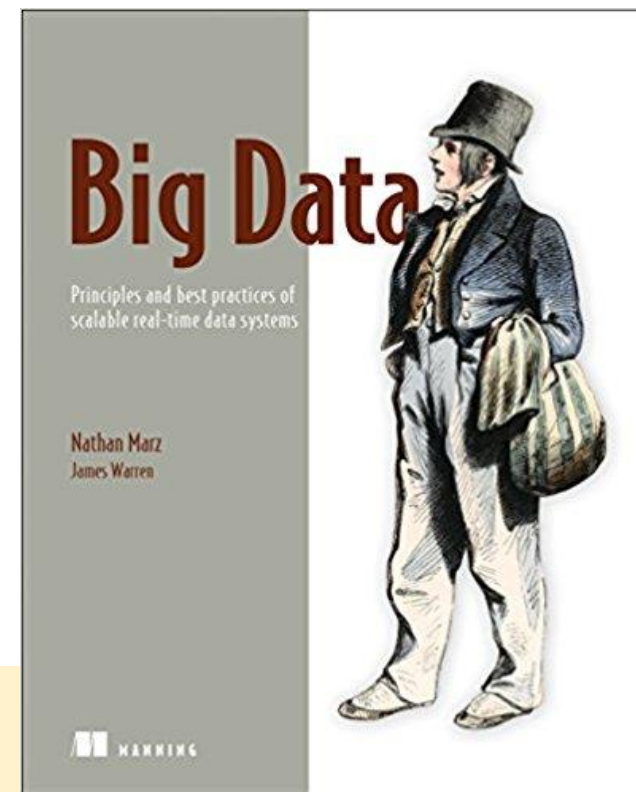


- Suggested video lectures
 - https://www.youtube.com/watch?reload=9&v=gIL8rW_eyww
 - <https://www.youtube.com/watch?v=llgU1UqL2JQ>



Big Data Architectures

Lambda Architecture



- Let's consider an example
 - Suppose your boss asks you to build a simple web analytics application. The application should track the number of pageviews to any URL a customer wishes to track. The customer's web page pings the application's web server with its URL everytime a pageview is received.
 - Additionally, the application should be able to tell you at any point what the top 100 URL's are by number of pageviews.
- ***Scaling issues with a traditional database.***

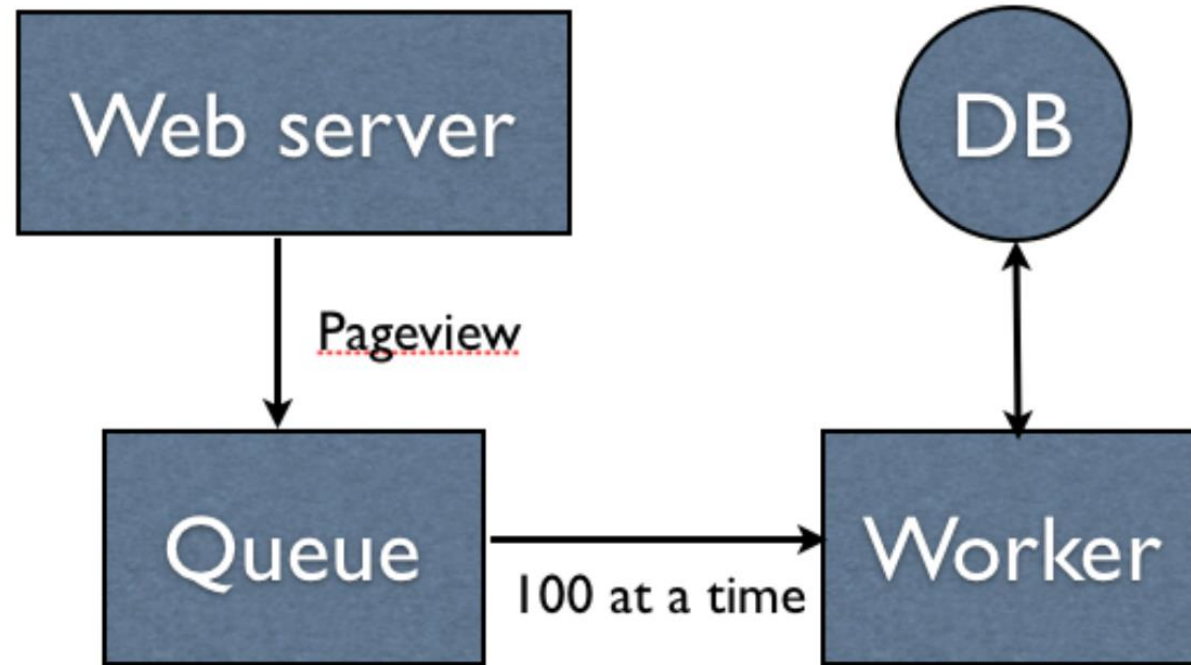
- The web analytics product is a huge success but you will incur very soon into the following problem

**TIMEOUT ERROR ON INSERTING
TO THE DATABASE!**

Limits of Traditional Tools



- **Scaling with a queue.** Resolves the timeout issues you were getting. It even has the added bonus that if the database ever gets overloaded again, the queue will just get bigger instead of timing out to the web server and potentially losing data.



Batching updates with queue and worker.

Limits of Traditional Tools



- ***Scaling by sharding the database.*** As the application gets more and more popular, you keep having to reshard the database into more shards to keep up with the write load. Each time gets more and more painful as there's so much more work to coordinate.

Limits of Traditional Tools



- ***Scaling by sharding the database.*** As the application gets more and more popular, you keep having to reshard the database into more shards to keep up with the write load. Each time gets more and more painful as there's so much more work to coordinate.
- ***Fault-tolerance issues begin.*** Eventually you have so many shards that it's not uncommon for the disk on one of the database machines to go bad. So that portion of the data is unavailable while that machine is down.

Limits of Traditional Tools



- ***Scaling by sharding the database.*** As the application gets more and more popular, you keep having to reshard the database into more shards to keep up with the write load. Each time gets more and more painful as there's so much more work to coordinate.
- ***Fault-tolerance issues begin.*** Eventually you have so many shards that it's not uncommon for the disk on one of the database machines to go bad. So that portion of the data is unavailable while that machine is down.
- ***Corruption issues.*** You don't notice until 24 hours later but by then the damage is done: many of the values in your database are inaccurate. Your system has no resilience to a human making a mistake.

Recap - Problems with traditional architecture



- *Fault-tolerance is hard*
- *Complexity pushed to application layer*
- *Lack of human fault-tolerance*
- *Maintenance is an enormous amount of work*

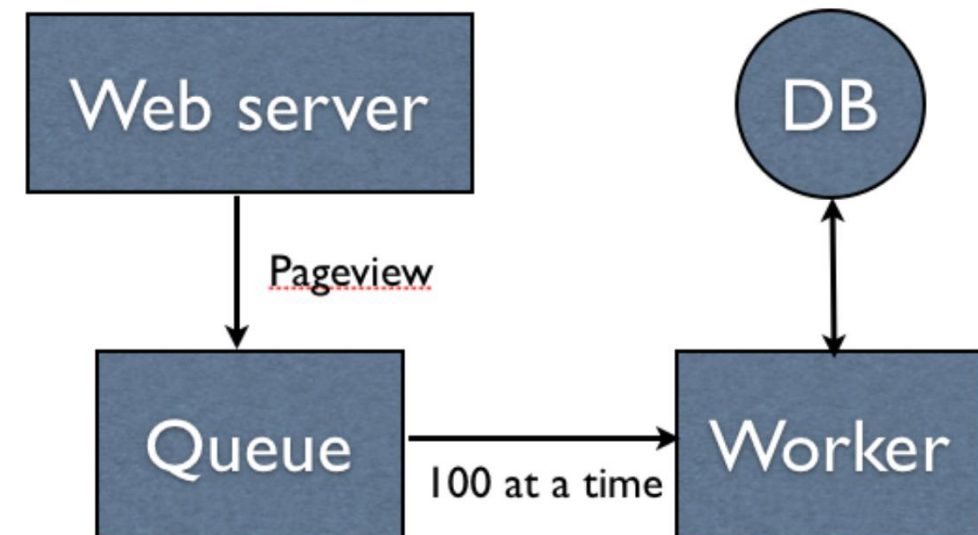
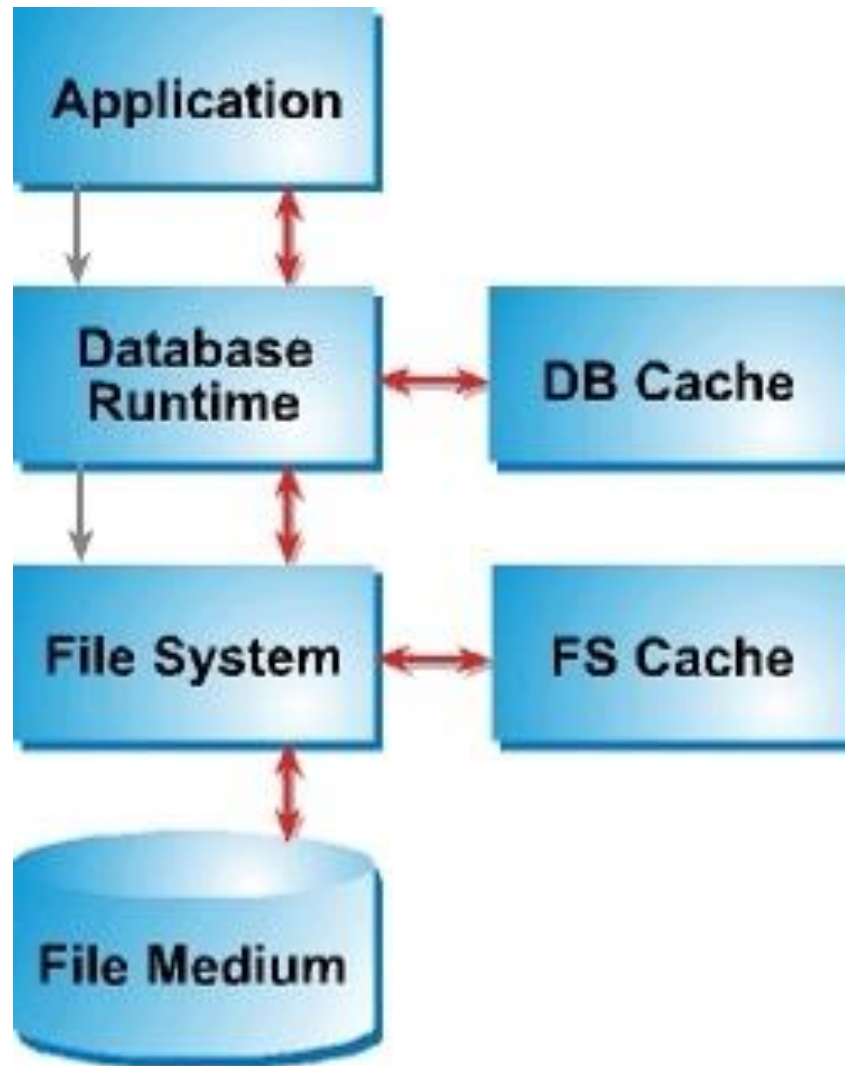
Recap - Problems with traditional architecture



- *Fault-tolerance is hard*
- *Complexity pushed to application layer*
- *Lack of human fault-tolerance*
- *Maintenance is an enormous amount of work*

**Solution: Big Data
Techniques & Technologies!!!**

Before Big Data Paradigm



- ... Traditional systems have failed to scale Big Data
 - Not Scalable
 - Too Complex
- The paradigm shifted to **Lambda Architecture**

Speed layer

Serving layer

Batch layer

What is a informational system?



- A data system answers **questions based on information** that was **acquired in the past**

Big Data Techniques & Technologies



- The databases and computation systems you use for Big Data are **self-aware of their distributed nature**.
- Another core technique you will learn about is **making your data immutable**.
- Big Data techniques **can scale to so much data**, you have the ability to design systems in different ways.
- Hadoop, for example, can **parallelize large scale batch computations** on very large amounts of data, but the computations have high latency.
- NoSQL databases like Cassandra achieve their scalability by offering you a much **more limited data model** than you're used to with something like SQL.

Properties of a Big Data Information System



- ***Robust and fault-tolerant.*** Immutability and recomputation into the core of your systems to make your systems innately resilient to human error.
- ***Low latency reads and updates.*** The vast majority of applications require reads to be satisfied with very low latency, typically between a few milliseconds to a few hundred milliseconds.
- ***Scalable.*** Scalability is the ability to maintain performance in the face of increasing data and/or load by adding resources to the system.
- ***General.*** Lambda Architecture generalizes to applications as diverse as financial management systems, social media analytics, scientific applications, and social networking.
- ***Extensible.*** Extensible systems allow functionality to be added with a minimal development cost.
- ***Minimal maintenance***
- ***Debuggable***

Batch Layer

- Stores the immutable constantly growing dataset
- Computes arbitrary views from this dataset using BigData technologies (can take hours)
- Can be always recreated

Serving Layer

- Indexing and exposing
 - the pre-computed batch views
 - the incremented real-time views
- Merges the batch and the real-time views

Speed Layer

- Computes views from the data stream in real-time
- Compensate for the high latency of the batch layer
- Incremental model and views are transient

Batch vs. Stream Processing



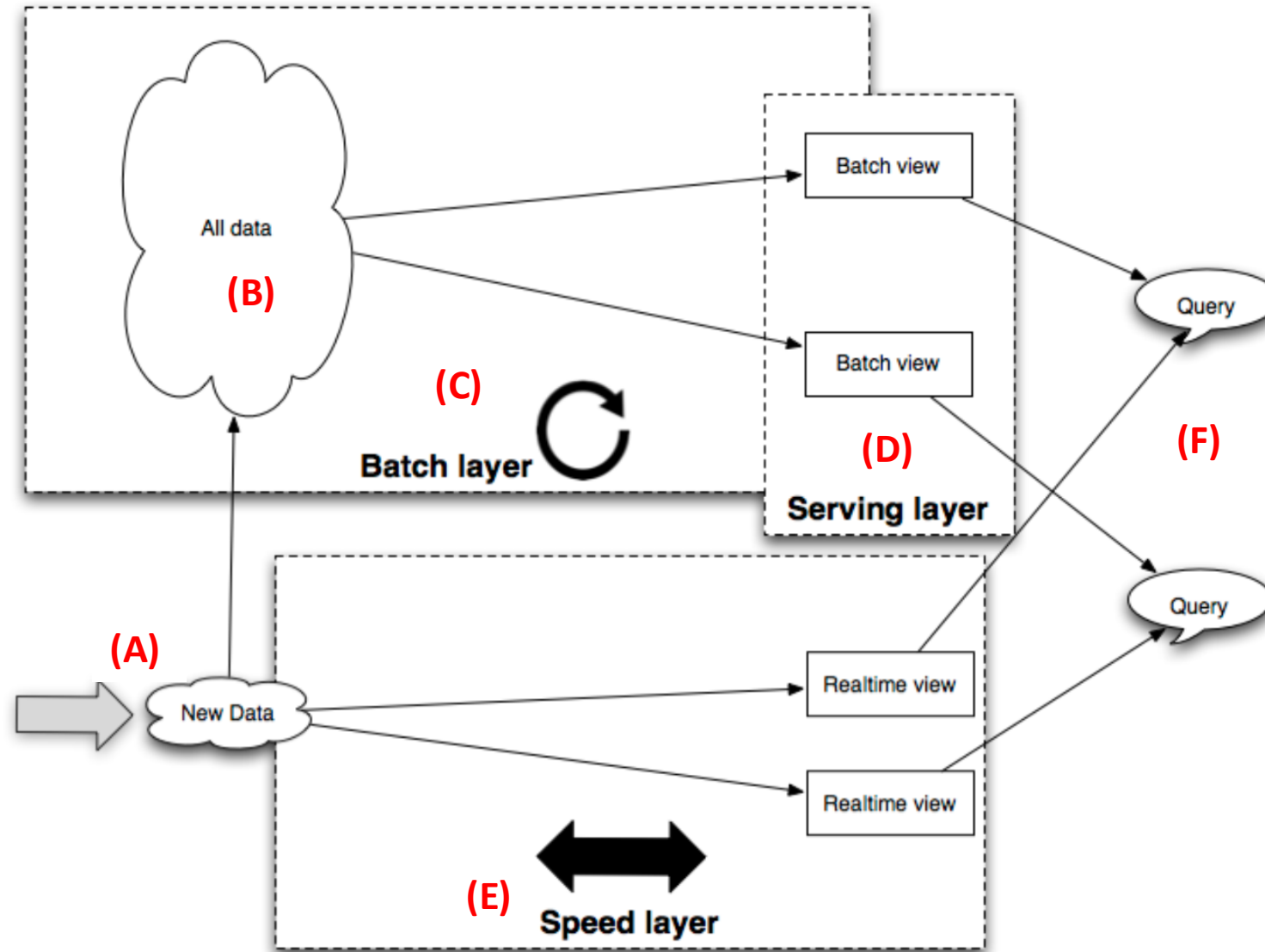
- Batch processing

- Gathering of data and processing as a group at one time.
- Jobs run to completion
- Data might be out of date

- Real-time processing

- Processing of data that takes place as the information is being entered.
- Run for ever

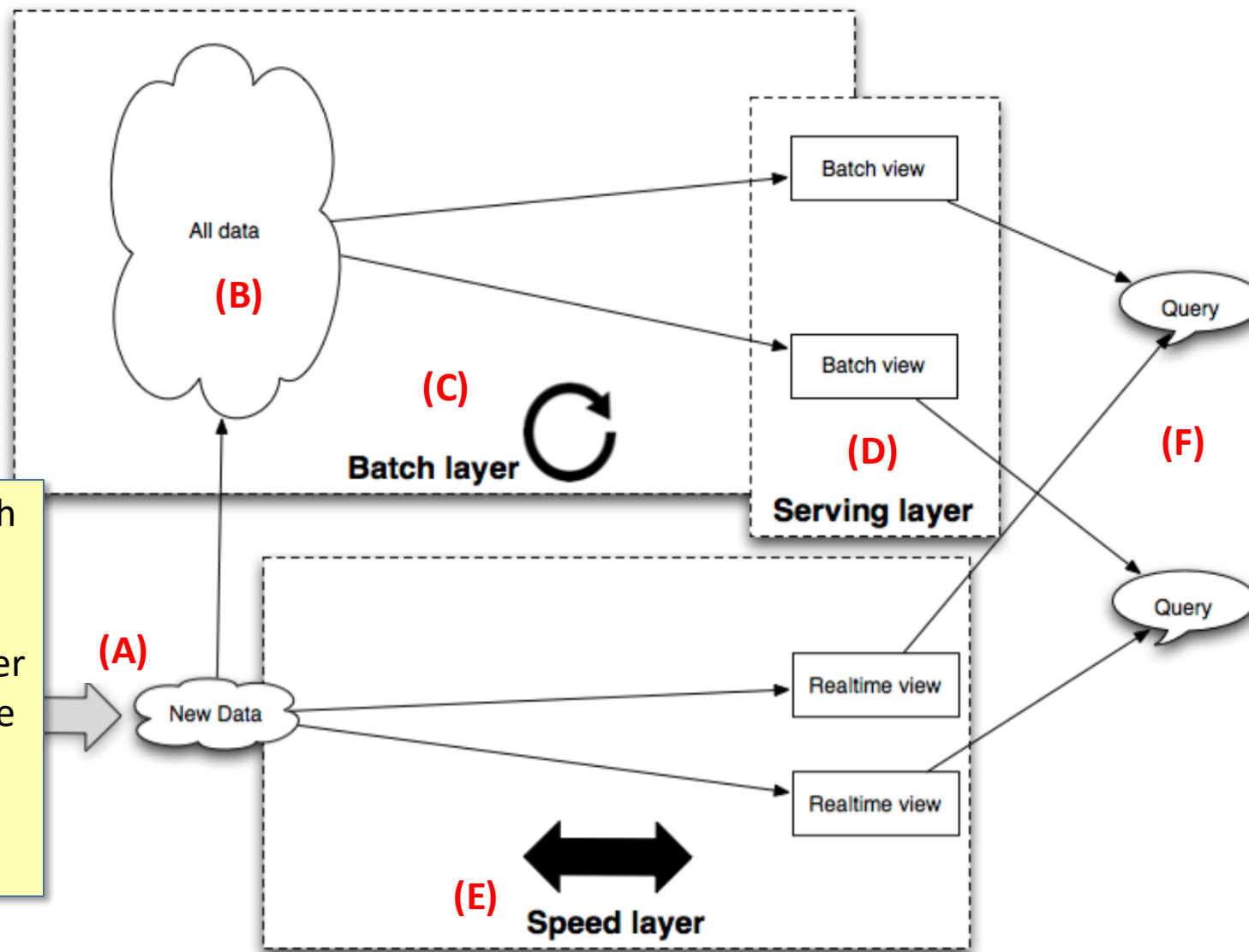
Lambda Architecture



Lambda Architecture



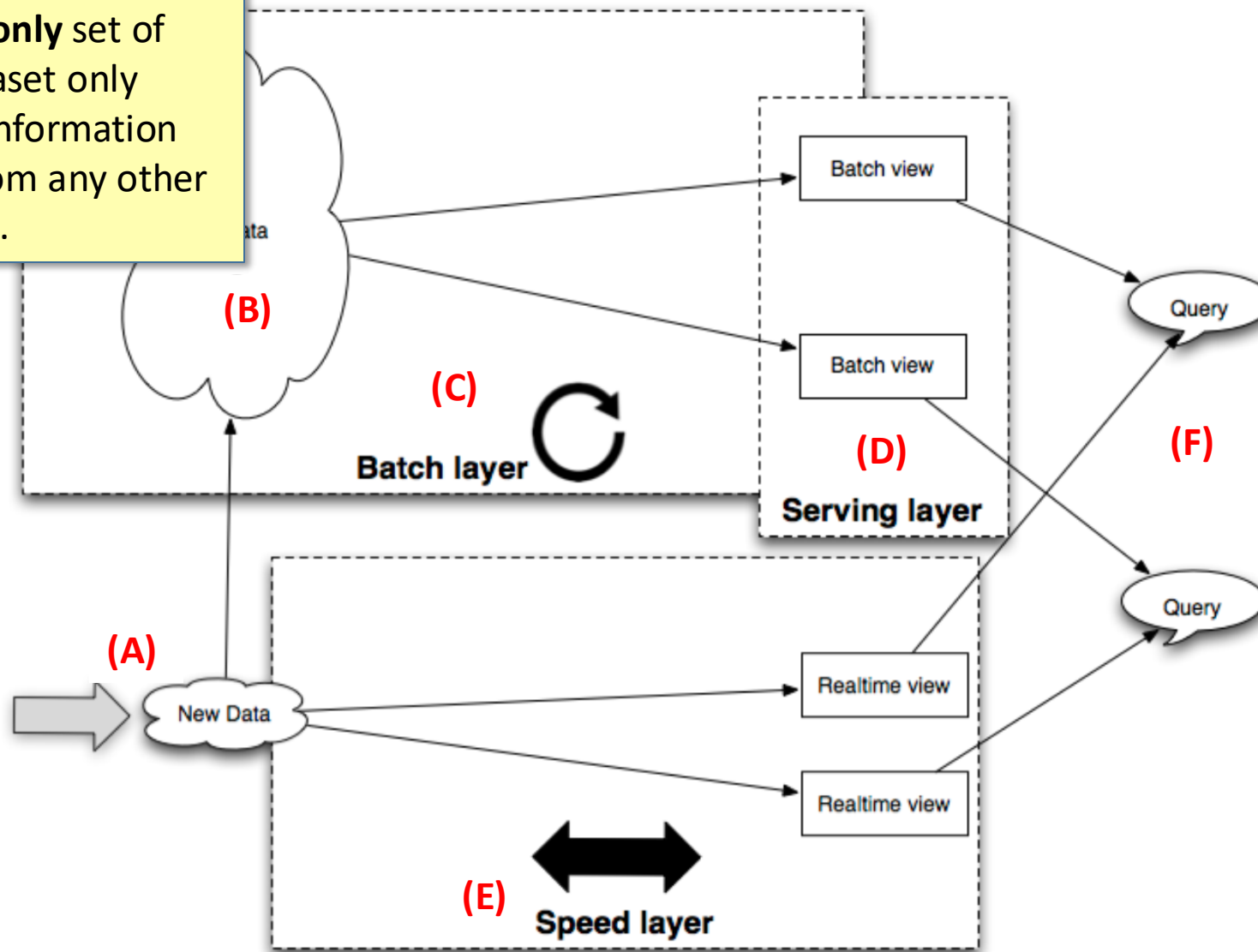
(A) All new data is sent to both the batch layer and the speed layer. In the batch layer, new data is appended to the master dataset. In the speed layer, the new data is consumed to do incremental updates of the realtime views.



Lambda Architecture



(B) The master dataset is an **immutable, append-only** set of data. The master dataset only contains the **rawest** information that is not derived from any other information you have.



Big Data Processing Technologies



Infrastructure

Hadoop On-Premise
cloudera
Hortonworks
MAPR
Pivotal
IBM InfoSphere
bluedata
jethro

Hadoop in the Cloud
amazon web services
Microsoft Azure
Google Cloud Platform
IBM InfoSphere
CAZENA
altiscale
Quale

Spark
databricks
GridGain
TACHYON
NEXUS

Cluster Services
amazon web services
kubernetes
docker
MESOSPHERE
Core OS
pepperdata
StackIQ

Analytics

Analyst Platforms
Palantir
AYASDI
Quid
enigma
Digital Reasoning
ORBITAL INSIGHT

Analytics Platforms
Microsoft
guavus
Datameer
Bottlenose
interlana

Data Science Platforms
context relevant
CONTINUUM
DataRobot
Alpine
MODE
plotly
ARIMO
dataiku
tonian
DOMINO
sense
yhat
ALGORITHMIA

Visualization
tableau
Google Cloud Platform
Qlik
looker
Roambi
Sisense
COMDATA
datorama
CHARTIO

Applications

Sales & Marketing
RADIUS
Gainsight
bloomreach
Zeta
EVERSTRING
livefyre
blueyonder
Lattice
kahuna
infer
SAILTHRU
persado
AVISO
sense
QUANTIFIND
ACTIONIQ
fuse|machines
ENGAGIO

Customer Service
MEDALLIA
ATTENITY
CLARABRIDGE
CLICKFOX
STELLA Service
NGDATA
Preact
DigitalGenius
appuri
Wise.io

Human Capital
gild
Connectifier
textio
entelo
hiQ

Legal
RAVEL
JUDICATA
Everlaw
Brevia
PREMONITION

NoSQL Databases
amazon DynamoDB
Google Cloud Platform
Microsoft Azure
mongoDB
MarkLogic
DATASTAX
Couchbase
SequoiaDB
redislabs
influxdata

NewSQL Databases
SAP HANA
Clustrix
Pivotal
paradigm4
nuODB
splice
MACHINE
citustdata
MariaDB
VOLTDB
deepdb
Trafection
Cockroach LABS

BI Platforms
Power BI
amazon web services
DOMO
Wave Analytics
GoodData
birst
kyvos insights
platforma
atscale
ARCADIA
SISENSE

Statistical Computing
sas
SPSS
MATLAB

Log Analytics
splunk
sumologic
kibana
CLOUD PHYSICS
loggly

Social Analytics
Hootsuite
NETBASE
DATASIFT
tracx
bitly
synthesio
simplereach

Ad Optimization
AppNexus
MediaMath
criteo
OpenX
rocketfuel
Integral
theTradeDesk
Ad Science
Adgorithms
dstillery
LiveIntent
DataXu
Appier
TAPAD
MOAT

Security
CYLANCE
CounterTack
cybereason
AREA 1 SECURITY
SentinelOne
Recorded Future
Guardian Analytics
FORTSCALE
sift science
Keybase
feedzai
SIGNIFYD

Vertical AI Applications
facebook
Clara
KASIST
lumina

Graph Databases
neo4j
STAPH
OrientDB
InfiniteGraph

MPP Databases
TERADATA
VERTICA
Microsoft Azure
Pivotal
lake
arks

Cloud EDW
amazon web services
Microsoft Azure
Pivotal
lake
arks

Data Transformation
alteryx
talend
TRIFACTA
tamr
Paxata
StreamSets
Alation

Data Integration
informatica
Put potential to work:
MuleSoft
snapLogic
Bedrock Data
xplenty

Real-Time
amazon web services
METAMARKETS
striim
confluent
DATATORRENT
dataArtisans

Machine Learning
Azure Machine Learning
amazon web services
H2O
SKYTREE
Dato
rapidminer
DATARPM
deepinsight
VIZEN
PredictionIO
glowfish

Speech & NLP
NarrativeScience
NUANCE
WolframAlpha
semantic machines
ARRIA
api.ai
Gridspace
cortical.io
malibu
MindMeld
iDIBON
yseop

Horizontal AI
IBM Watson
Cortana
sentient
vicarious
nervana
Numenta
HyperScience
SI
clarifai
Dextro
MetaMind

Publisher Tools
Outbrain
Taboola
quantcast
Chartbeat
yieldbot
Yieldmo

Govt / Regulation
Socrata
OPENGOV
FN
FiscalNote
enigma
mark43
PREDPOL
OpenDataSoft

Finance
affirm
LendingClub
OnDeck
Kreditech
zest finance
LendUp
Kabbage
tidemark
INSIKT
zuora
Dataminr
Lenddo
KENSHC
AIDYIA
iSENTIUM
Quantopian

Management / Monitoring
New Relic
APDDYNAMICS
amazon
CODE42

Storage
amazon web services
Google Cloud Platform
Microsoft Azure
age

App Dev
apigee
CASK
Keen IO
Typesafe
DRIVEN

Crowd-sourcing
amazon mechanical turk
CrowdFlower
WorkFusion

Search
hp
Autonomy
ORACLE
ENDECA
EXALEAD
Lucidworks
elastic
ThoughtSpot
MAANA
swiftype
Algolia
SINEQUA

Data Services
UC OPERA
Mu Sigma
EXL
DATA SCIENCE
kaggle
dataSCOPE
DataKind

For Business Analysts
OrigamiLogic
ClearStory
CIRRO
import io

Web / Mobile / Commerce
Google Analytics
mixpanel
RJMetics
BLUECORE
AMPLITUDE
granify
sumail
Airtable
retention
custora

Education / Learning
KNEWTON
Clever
eclara
PANORAMA
knowte

Life Sciences
23andMe
Counsyl
Recombine
KYRUUS
FLATIRON
zymergen
HealthTap
METABIOTA
ZEPHYR
HEALTH
ovia
Ginger.io
transcriptic
Glow
enlilic
AiCure
Atomwise

Industries
eHarmony
OPower
RetailNext
STITCH FIX
WorkFusion
BLUE RIVER
TACHYUS
Seeq
FarmLogs
SwiftKey
HowGood
select
BOXXEVER

Cross-Infrastructure/Analytics

amazon web services Google Microsoft SAP SAS 1010 hp Antares VERTICA vmware TIBCO TERADATA ORACLE NetApp



Framework
hadoop HDFS
YARN
Spark
MESOS
TEZ
Flink
CDAP

Coordination
talend
Apache Zookeeper
Apache Ambari

Real-Time
STORM
Spark
APEX
Flink
TACHYON
druid

Stat Tools
ScalaLab
NumPy
SciPy

Machine Learning
milib
Aerosolve
Caffe
learn
WEKA
VELES
SINGA
MADlib
CNTK
TensorFlow
FeatureFu
DIMSUM
jupyter
DL4J

Search
elasticsearch
Solr
Lucene

Security
Apache Ranger
Zeppelin

Open Source

Health
JAWBONE GARMIN
practicefusion
fitbit
VALIDIC
netatmo
kinsa
Human API

IoT
UPTAKE
helium
samara
ThingWorx
samsara
estimate

Financial & Economic Data
Bloomberg
DOW JONES
THOMSON REUTERS
YODLEE
PREMISE
S&P CAPITAL IQ
quandl
xignite
CB INSIGHTS
mattermark
StockTwits
estimote
PLAID

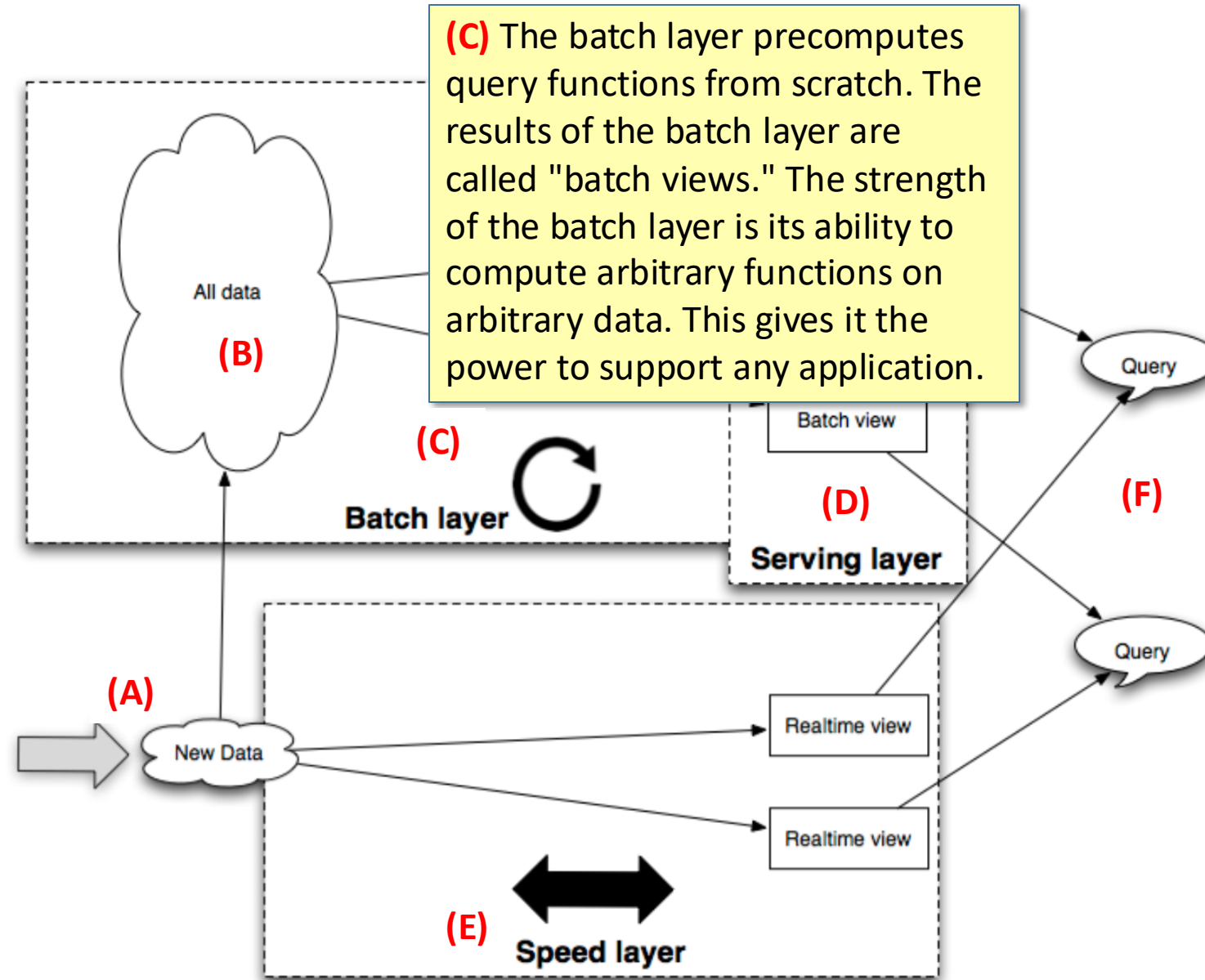
Air / Space / Sea
PLANET LABS
spire
WINDWARD
CRUISE
SKYWATCH
Airware
DroneDeploy

Location / People / Entities
axiom
Experian
EPSILON
InsideView
GARMIN
foursquare
STREETLINE
Crimson Hexagon
CARTO DB
factual
PlaceIQ
CIRCULATE
placemeter
BASIS
Sense

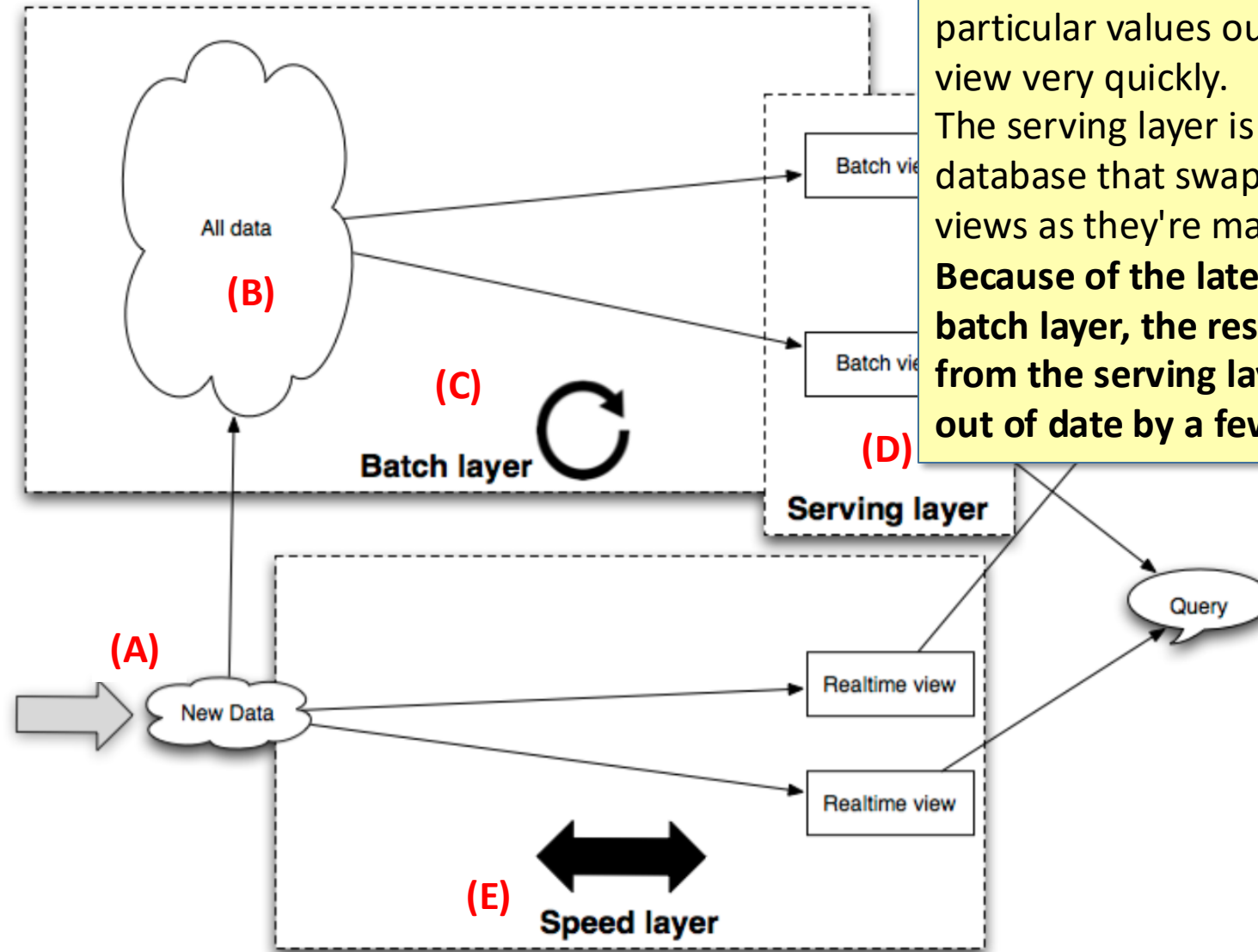
Other
qualtrics
panjiva
DATA.GOV

Incubators & Schools
GA
PLURALSIGHT
INSIGHT
DataCamp
DataElite
The Data Incubator
METIS

Lambda Architecture



Lambda Architecture



(D) The serving layer indexes the batch views produced by the batch layer and makes it possible to get particular values out of a batch view very quickly. The serving layer is a scalable database that swaps in new batch views as they're made available. **Because of the latency of the batch layer, the results available from the serving layer are always out of date by a few hours.**

Big Data Storage Technologies



Infrastructure

Analytics

Applications

Infrastructure

- Hadoop On-Premise:** cloudera, Hortonworks, MAPR, Pivotal, IBM InfoSphere, bluedata, jethro
- Hadoop in the Cloud:** amazon web services, Microsoft Azure, Google Cloud Platform, IBM InfoSphere, CAZENA, TREASURE DATA, altiscale, Qu bale
- Spark:** databricks, GridGain, TACHYON NEXUS
- Cluster Services:** amazon web services, kubernetes, docker, HPCC SYSTEMS, MESOSPHERE, Core OS, pepperdata, StackIQ
- Analyst Platforms:** Palantir, AYASDI, Quid, enigma, Digital Reasoning, ORBITAL INSIGHT
- Analytics Platforms:** Microsoft guavus, Datameer, Bottlenose, interlana
- Data Science Platforms:** context relevant, CONTINUUM, DataRobot, Alpine, MODE, plotly, ARIMO, dataiku, tonian, DOMINO, sense, yhat, ALGORITHMIA
- Visualization:** tableau, Google Cloud Platform, Qlik, Looker, Roambi, Sisense, COMDATA, datorama, CHARTIO
- Sales & Marketing:** RADIUS, Gainsight, bloomreach, Zeta, EVERSTRING, livefyre, blueyonder, Lattice, kahuna, infer, SAILTHRU, persado, AVISO, sense, QUANTIFIND, ACTIONIQ, fuse|machines, ENGAGIO
- Customer Service:** MEDALLIA, ATTENITY, CLARABRIDGE, CLICKFOX, STELLA Service, NGDATA, Preact, DigitalGenius, appurri, Wise.io
- Human Capital:** gild, Connectifier, textio, entelo, hiQ
- Legal:** RAVEL, JUDICATA, Everlaw, Brevia, PREMONITION
- NoSQL Databases:** amazon DynamoDB, Google Cloud Platform, ORACLE, Microsoft Azure, mongoDB, MarkLogic, DATASTAX, Couchbase, SequoiaDB, redislabs, influxdata
- NewSQL Databases:** SAP HANA, Clustrix, Pivotal, paradigm4, nuODB, splice machine, citusdata, MariaDB, VOLTDB, cockroach LABS, deepdb, Trafodion
- BI Platforms:** Power BI, amazon web services, Domo, Wave Analytics, birst, GoodData, kyvros insights, platforma, atscale, ARCADIA, SISENSE
- Statistical Computing:** sas, SPSS, MATLAB
- Log Analytics:** splunk, sumologic, kibana, CLOUD PHYSICS, loggly
- Social Analytics:** Hootsuite, NETBASE, DATASIFT, tracx, bitly, synthesio, simplereach
- Ad Optimization:** AppNexus, MediaMath, critico, OpenX, rocketfuel, Integral, theTradeDesk, AdScience, Adgorithms, distillery, LiveIntent, DataXu, Appier, TAPAD, MOAT
- Security:** CYLANCE, CounterTack, cybereason, AREA 1 SECURITY, SentinelOne, Recorded Future, Guardian Analytics, FORTSCALE, sift science, Keybase, feedzai, SIGNIFYD
- Vertical AI Applications:** facebook, Clara, KASIST, lumia
- Graph Databases:** neo4j, STAPH, OrientDB, InfiniteGraph
- MPP Databases:** TERADATA, VERTICA, Netezza, Qlik, Action, cognito, EXASOL, dremio
- Cloud EDW:** amazon web services, Google Cloud Platform, Microsoft Azure, Pivotal, snowflake, WATERLINE DATA, Infoworks
- Data Transformation:** alteryx, talend, TRIFACTA, tamr, Paxata, StreamSets, Alation
- Data Integration:** informatica, Put potential to work, MuleSoft, snapLogic, Bedrock Data, xplenty
- Real-Time:** amazon web services, METAMARKETS, striim, confluent, DATATORRENT, dataArtisans
- Machine Learning:** Azure Machine Learning, amazon web services, H2O, SKYTREE, Dato, rapidminer, DATARPM, deepsense, VIZEN, PredictionIO, glowfish
- Speech & NLP:** NarrativeScience, NUANCE, WolframAlpha, semantic machines, ARRIA, api.ai, Gridspace, cortical.io, maluba, MindMeld, iBIBON, yseop
- Horizontal AI:** IBM Watson, Cortana, sentient, viv, nervana, vicarious, nana, Numenta, HyperScience, SI, clarifai, Dextro, MetaMind
- Publisher Tools:** Outbrain, Taboola, quantcast, Chartbeat, yieldbot, Yieldmo
- Govt / Regulation:** Socrata, OPENGOV, FN, FiscalNote, enigma, mark43, PREDPOL, OpenDataSoft
- Finance:** Affirm, LendingClub, OnDeck, Kreditech, zest finance, LendUp, Kabbage, tidemark, Ruffly, INSIKT, Zuora, Dataminr, Lenddo, KENSHC, MYA, iSENTIUM, Quesada, iSentient
- Management / Monitoring:** New Relic, APPDYNAMICS, amazon web services, octifio, Numerify, splunk, DATADOG, TROCAN, DRIVEN, Anodot
- Security:** TANIUM, Illumio, CODE42, DataGravity, CipherCloud, VECTRA, sqrrl, BlueTalon
- Storage:** amazon web services, Google Cloud Platform, Microsoft Azure, panasas, nimblestorage, COHO, Qumulo
- App Dev:** apigee, CASK, Keen IO, Typesafe, DRIVEN
- Crowd-sourcing:** amazon mechanical turk, CrowdFlower, WorkFusion
- Search:** hp, Autonomy, ORACLE, ENDECA, EXALEAD, Lucidworks, elastic, ThoughtSpot, MAANA, swifttype, Algolia, SINEQUA
- Data Services:** UC OPERA, Mu Sigma, EXL, DATA SCIENCE, kaggle, datascope, DataKind
- For Business Analysts:** OrigamiLogic, ClearStory, CIRRO, import io
- Web / Mobile / Commerce:** Google Analytics, mixpanel, RUMetrics, MICROCORE, AMP, AMP, SUMA, rete, RUMetrics, MICROCORE, AMP, AMP, SUMA, rete

Analytics

Applications

Cross-Infrastructure/Analytics

amazon web services, Google, Microsoft, IBM, SAP, SAS, data, hp, Autonomy, VERTICA, vmware, TIBCO, TERADATA, ORACLE, Facebook, NetApp



Framework: hadoop, HADOOP, YARN, Spark, MESOS, TEZ, Flink, CDAP

Query / Data Flow: SLAMDATA, HIVE, DRILL, Google Cloud Dataflow

Data Access: cassandra, HBASE, mongoDB, CouchDB, riak, OPENTSDB, nifi

Coordination: talend, Apache Zookeeper, Apache Ambari

Real-Time: STORM, Spark, APEX, Flink, TACHYON, druid

Stat Tools: ScalaLab, NumPy, SciPy

Machine Learning: milib, Aerosolve, Caffe, WEKA, DIMSUM

Security: Apache Ranger, Zeppelin

Visualization: Tableau, Qlik, Looker, Roambi, Sisense, COMDATA, datorama, CHARTIO

Health: JAWBONE, GARMIN, practicefusion, fitbit, Withings, VALIDIC, netatmo, kinsa, Human API

IOT: UPTAKE, ThingWorx, helium, samsara, estimate

Financial & Economic Data: Bloomberg, DOW JONES, THOMSON REUTERS, YODLEE, PREMISE, S&P CAPITAL IQ, quandl, xignite, CB INSIGHTS, mattermark, StockTwits, estimate, PLAID

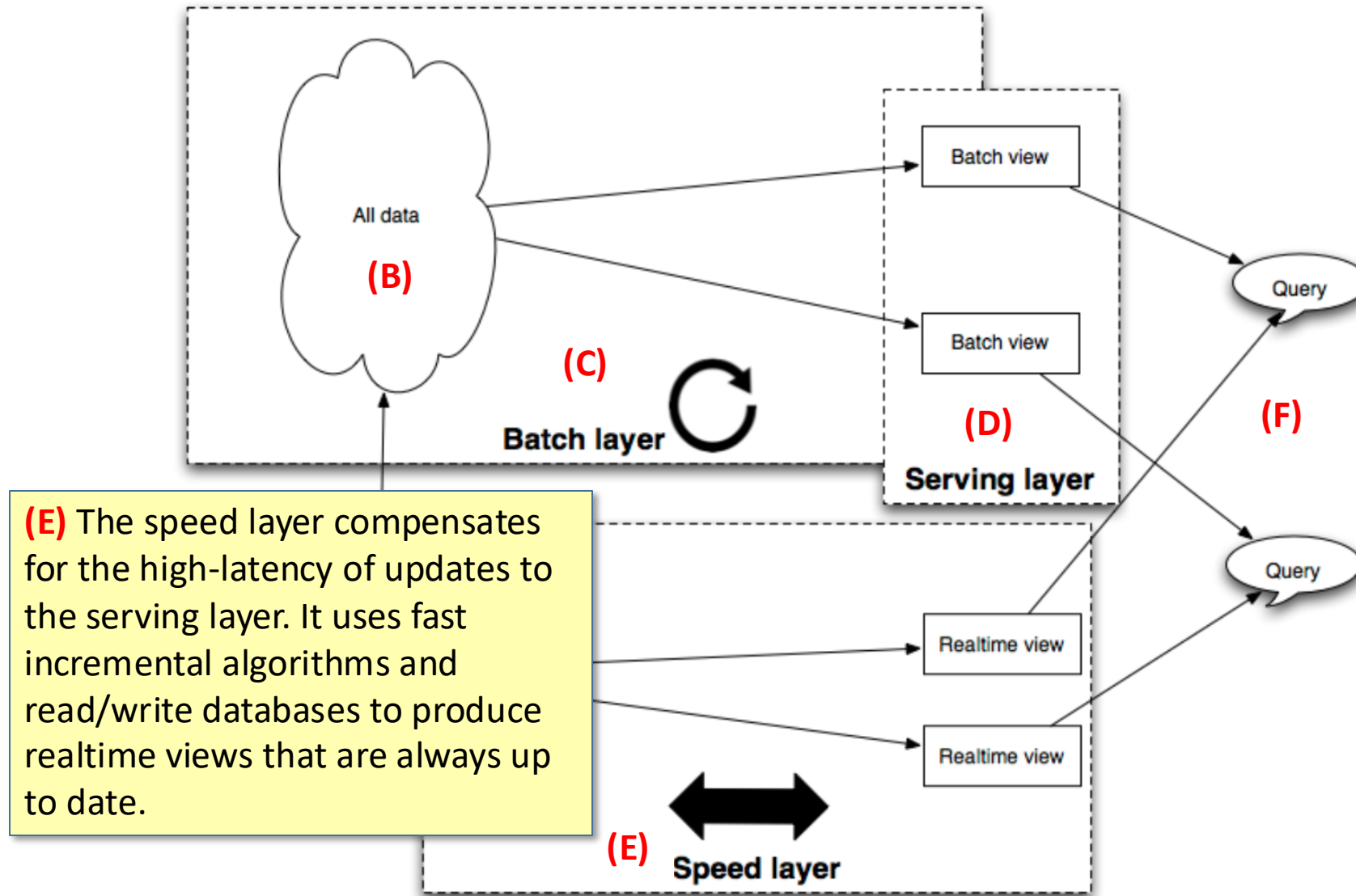
Air / Space / Sea: PLANET LABS, spire, WINDWARD, CRUISE, SKYWATCH, Airware, DroneDeploy

Location / People / Entities: axiomatic, Experian, EPSILON, InsideView, GARMIN, foursquare, STREETLINE, esri, Crimson Hexagon, CARTO, factual, PlaceIQ, CIRCULATE, placemeter, BASIS, Sense

Open Source: neo4j, cassandra

Security: CouchDB, relax, PLURALSIGHT, INSIGHT, DataElite, The Data Incubator, METIS

Lambda Architecture



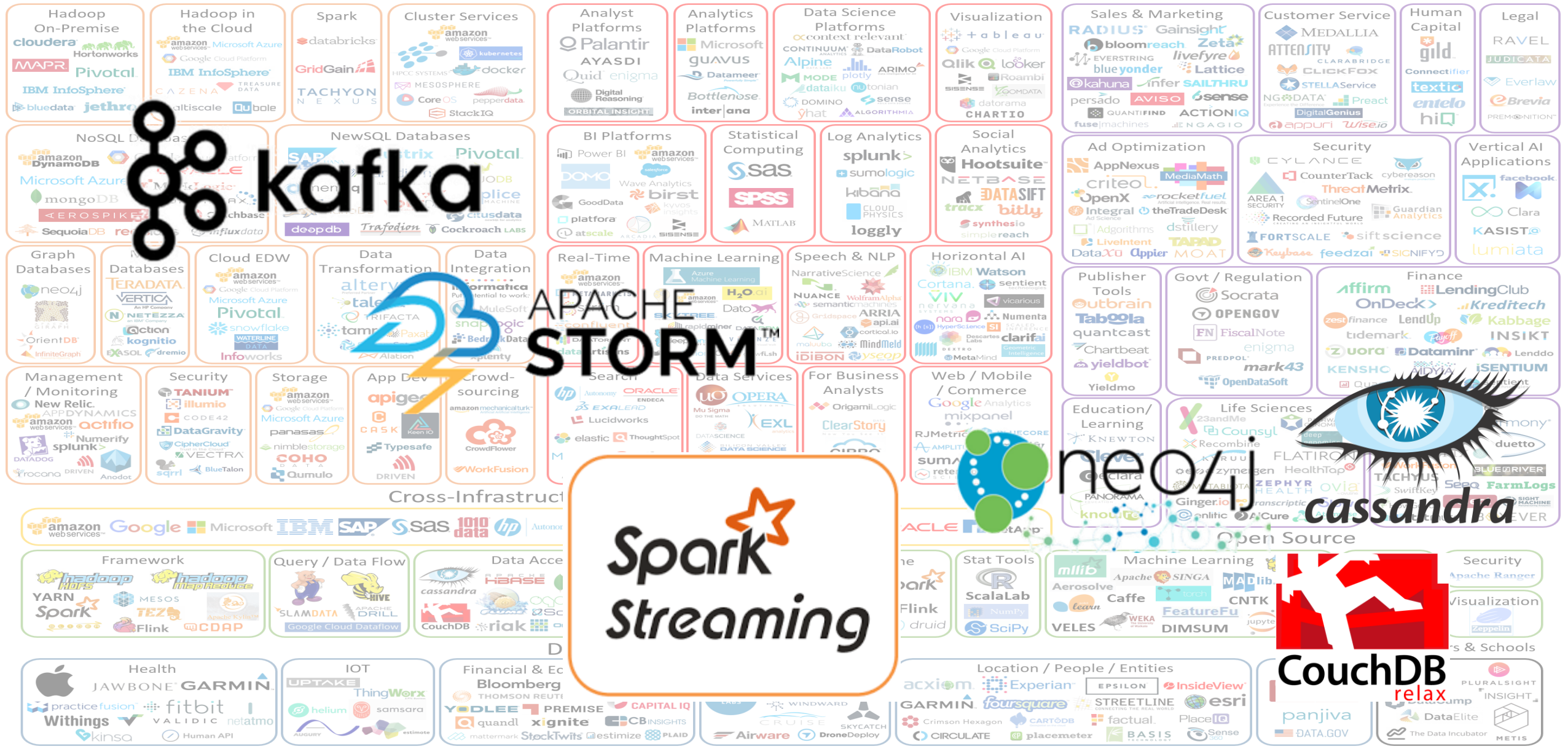


Big Data Processing Technologies

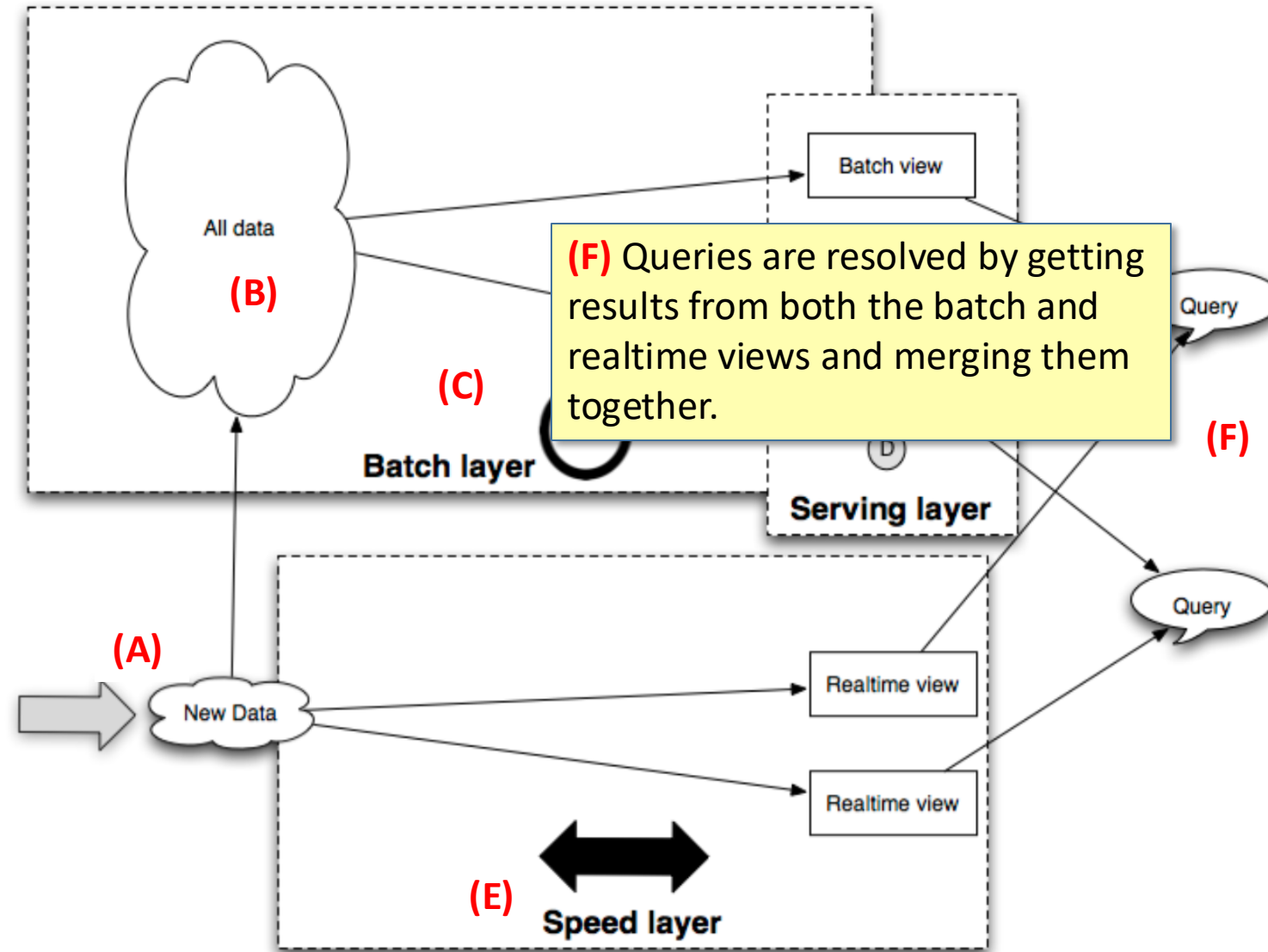
Infrastructure

Analytics

Applications



Lambda Architecture



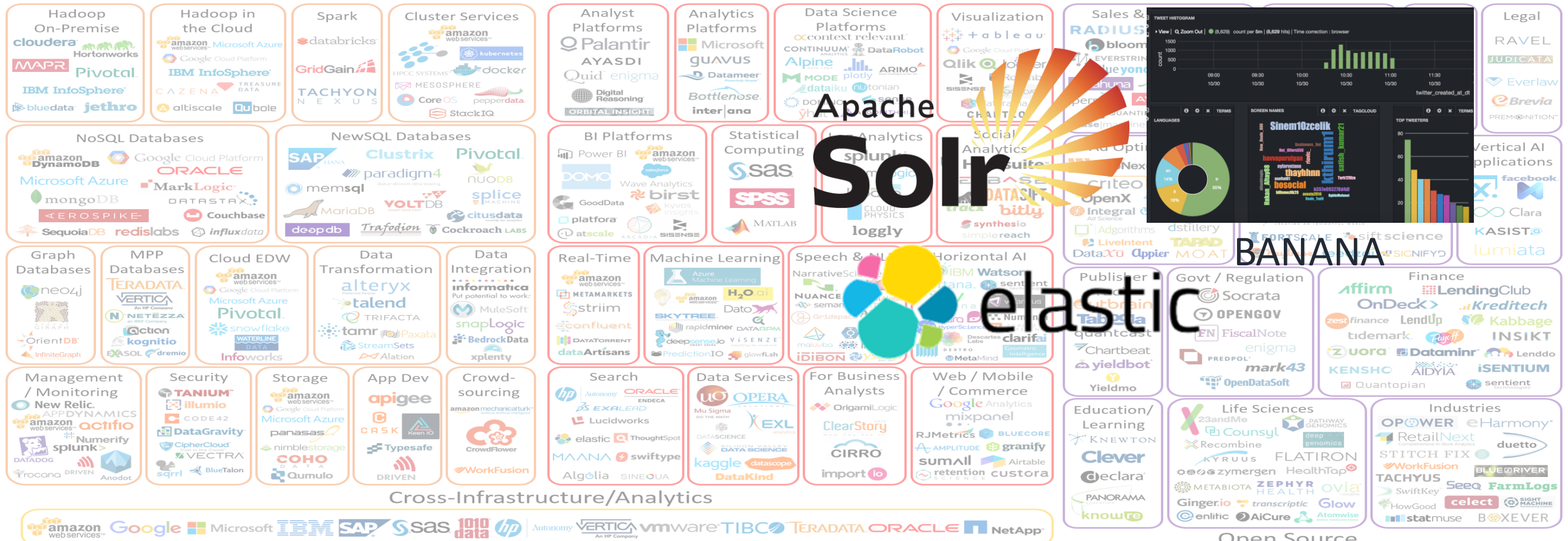
Big Data Storage/Searching Technologies



Infrastructure

Analytics

Applications



Cross-Infrastructure/Analytics

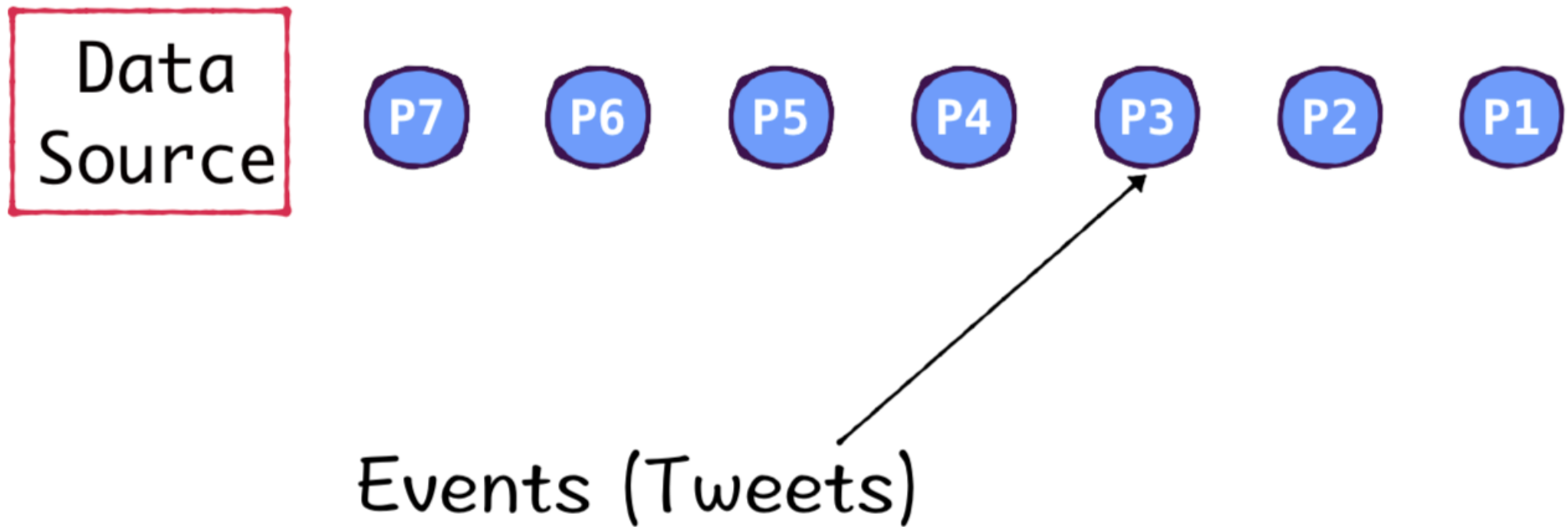
Open Source

Data Sources & APIs

Incubators & Schools



Batch vs Stream Processing



Batch vs Stream Processing



Batch vs Stream Processing



Gap increased during processing time



Batch vs Stream Processing



If processing time $\gg$ event arrival rate



Batch vs Stream Processing



How do we reduce processing time?



- Plethora of emerging technologies 4 Stream Processing
 - <https://thenewstack.io/apache-streaming-projects-exploratory-guide/>
- Apache Spark
- Apache Storm
- Apache Kafka Stream
- Apache Flink
- Apache Flume
- Apache S4
- Apache NiFi
- ...

Stream Processing



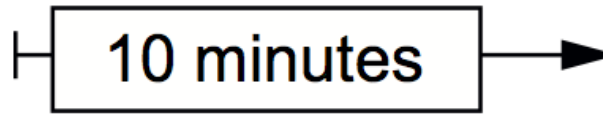
- Micro-batched
 - Spark streaming
 - Trident
- One-at-a-time
 - **Spark streaming**
 - Apache Storm
 - Apache Kafka
 - Alerting
 - Financial trading
 - ...

	One-at-a-time	Micro-batched
Lower latency	✓	
Higher throughput		✓
At-least-once semantics	✓	✓
Exactly-once semantics	In some cases	✓
Simpler programming model	✓	

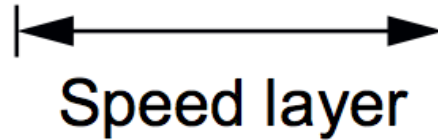
Expiring realtime views



Serving
layer

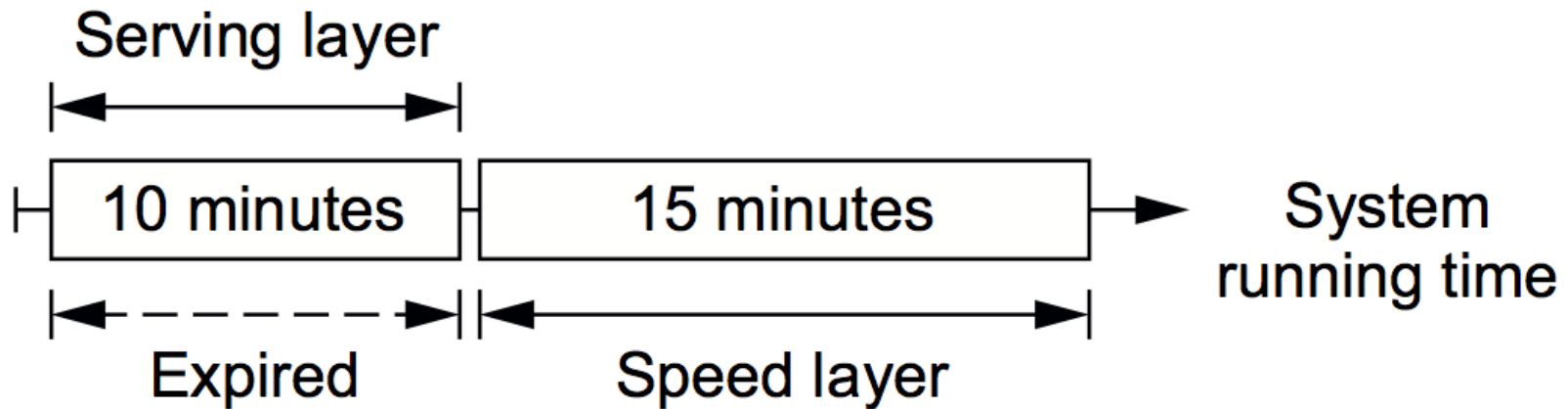


System
running time

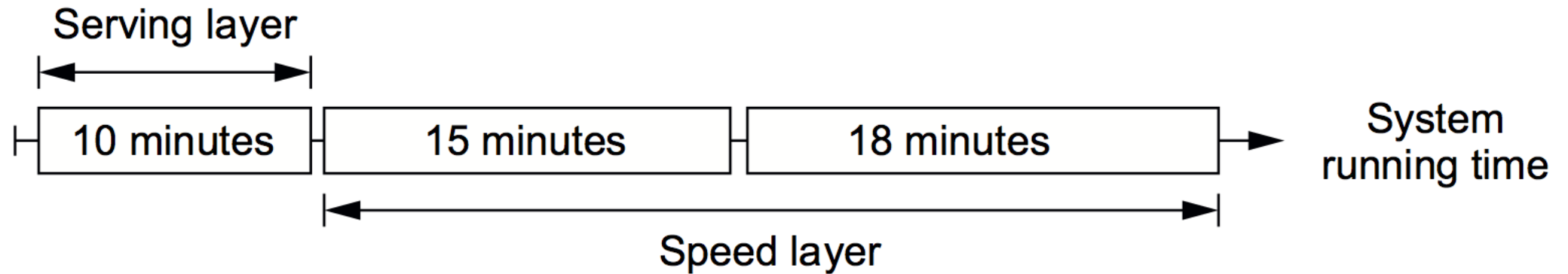


After the first batch computation run,
the serving layer remains empty but the
speed layer has processed recent data.

Expiring realtime views

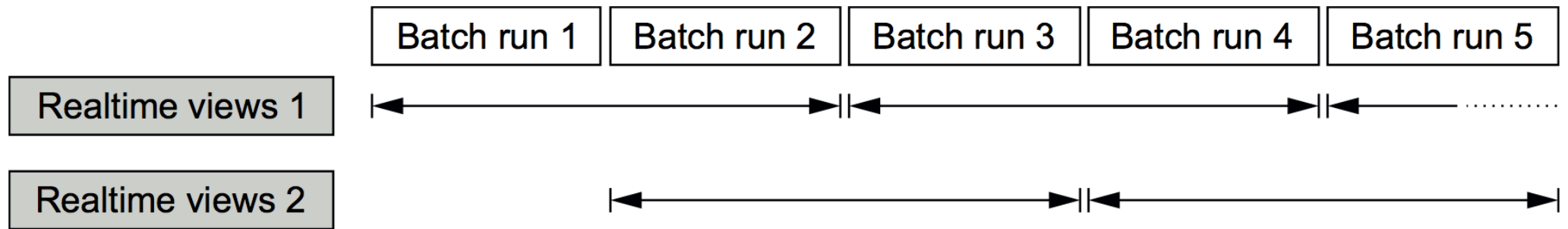


When the second batch computation run finishes, the first segment of data has been absorbed into the serving layer and can be expired from the speed layer views.

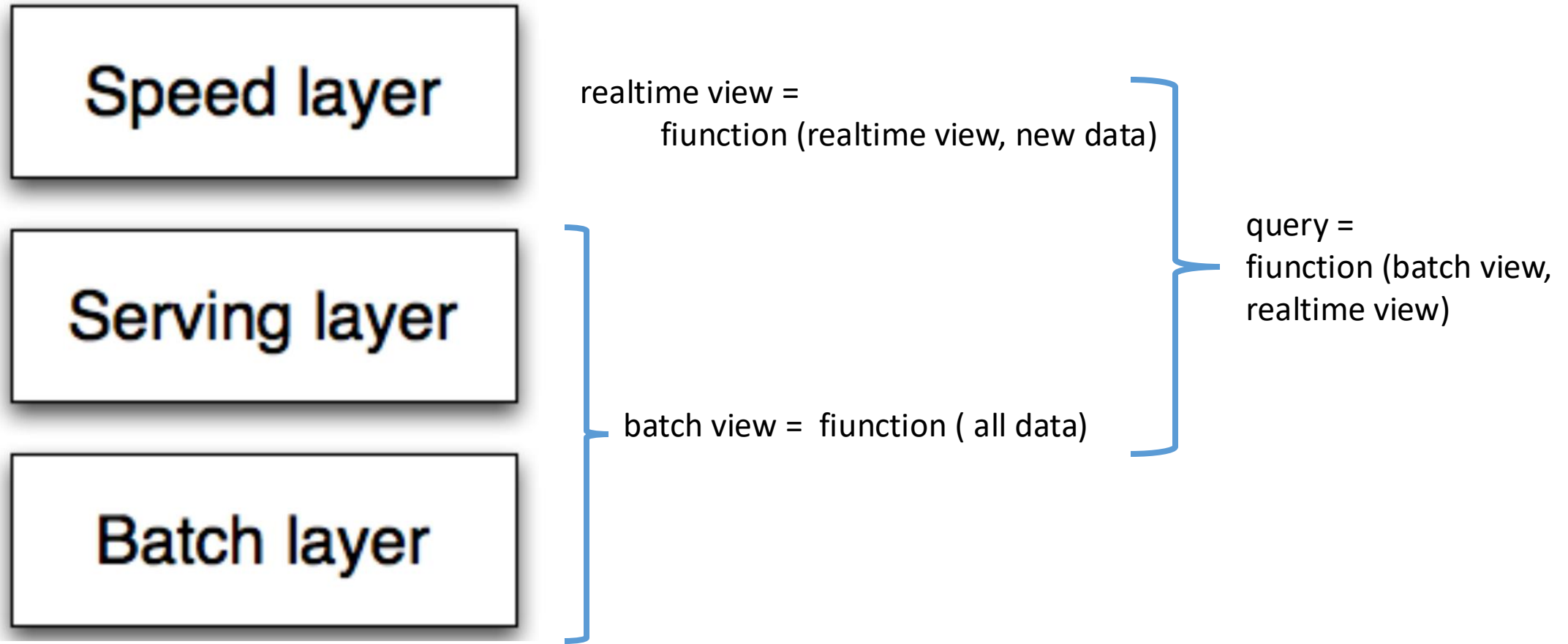


Just prior to the completion of the batch layer computation, the speed layer is responsible for data that accumulated during the prior two runs.

Expiring realtime views



Lambda Architecture



Traditional Approach



- Enterprise systems generally have one or more large relational databases at their center. These databases hold “the current state” of the system, and other applications query and update this state. However this approach has problems with race-conditions and performance bottlenecks.
- Sometimes such systems keep “history” information (aka audit trails) which theoretically show how the current state has been reached.



- For the reflection: Questioning the Lambda Architecture

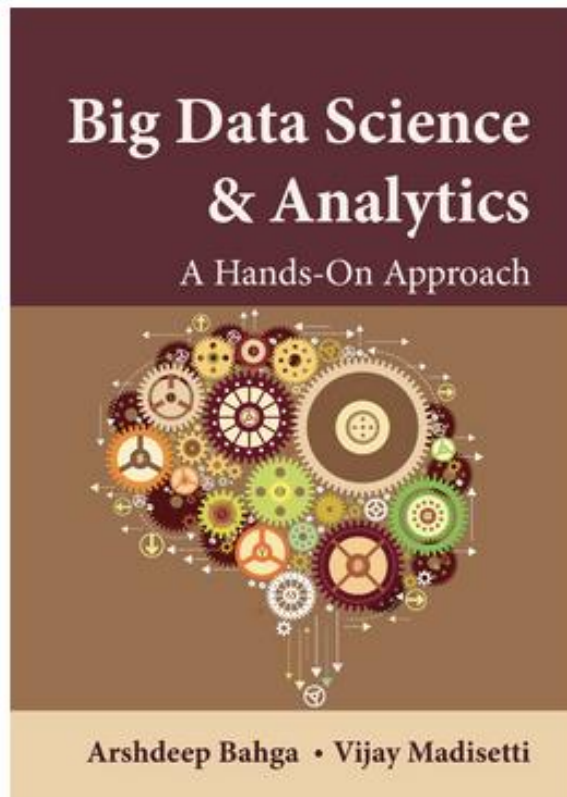
<https://www.oreilly.com/ideas/questioning-the-lambda-architecture>

<http://milinda.pathirage.org/kappa-architecture.com/>

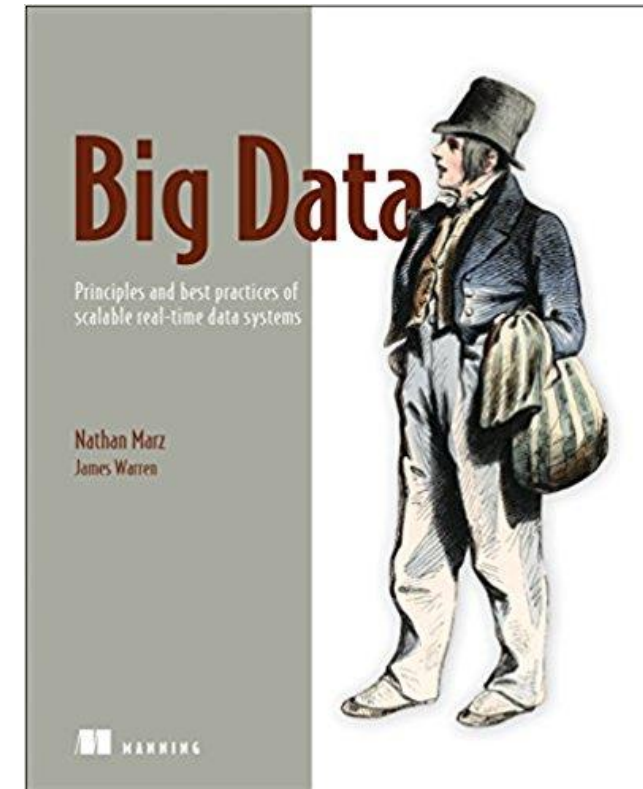
Suggested Readings



Read Chapter 1



Read Chapter 1





Project Works

"Requirements"



- **Data Ingestion:**

- Apache Airflow, Apache NiFi, Streaming API, MQTT, ...
- Public dataset file (for a group of 2 people)

- **Data Storage**

- HDFS, Big Data File Formats, NoSQL DB, ...

- **Data Processing**

- Batch and Interactive Query:
 - ❖ Apache Spark, Pig, HIVE, ...
- Real-time Analytics: at least one query (for group > 2 students)
 - ❖ Spark Streaming, Storm, KSQL, ...

- **Data Visualization**

- Apache Solr + BANANA, Elasticsearch + KIBANA, Zeppelin, D3.js, React, ...

How to collect data



- Datasets
- Endpoints
- API
- Scraping
- MQTT
- ...

Some Ideas



- IoT Analytics: sensors, smartphone, raspberry router, etc.
- Social Media: Twitter Streaming API, ...
- Web Analytics: By Scraping Booking.com, Subito.it, ...
- Epidemiological and Genomics Analytics: COVID19 dataset are available
- Cyber Intelligence: Open Source Intelligence
- ...

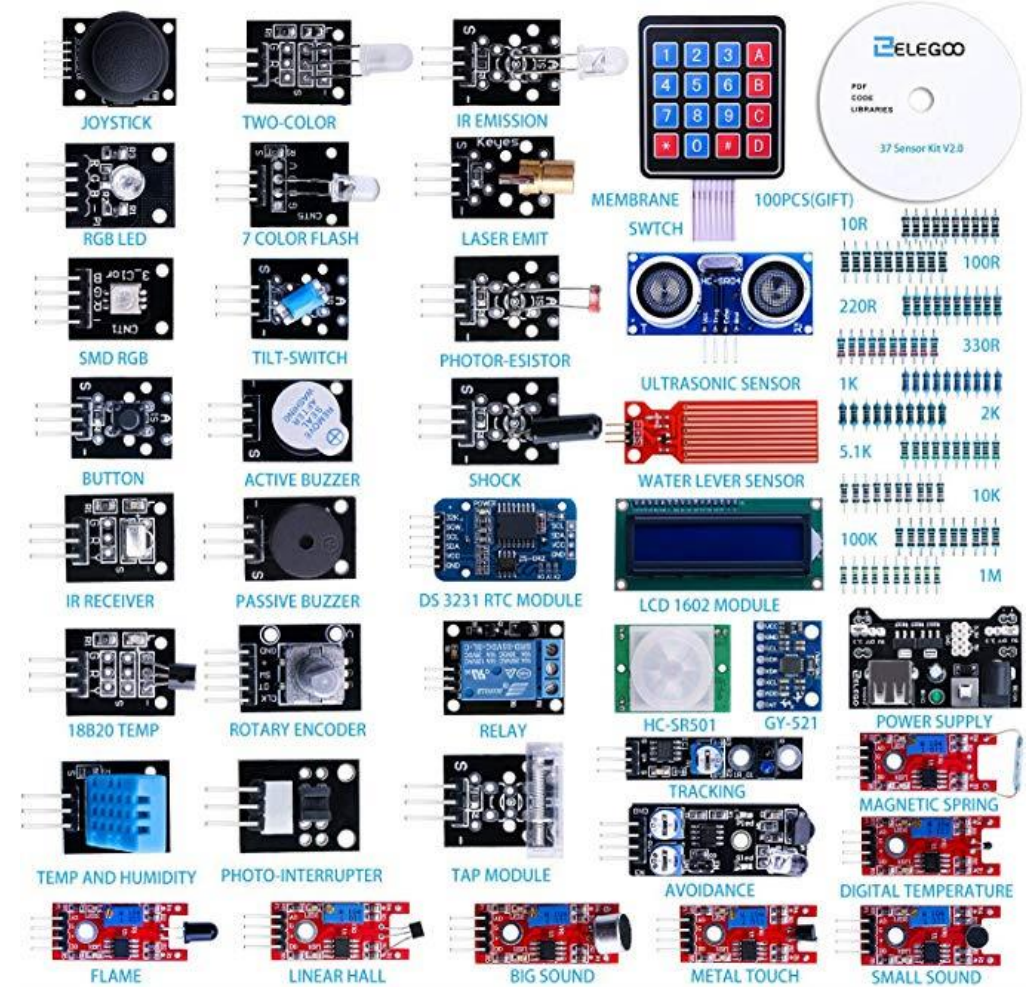
Smartphone Sensors



- Let's try to collect data from the smartphone, in order to perform some analytics about my activities and show them into an interactive dashboard
- **Data Ingestion:** <https://whatwebcando.today/device-position.html>
- **Data Storage:** choose and use the data file format, choose the repository (HDFS, KeyValue store, and so forth)
- **Data Processing:** let us assign some analytics about my activities, pre-compute and then show the results in a live dashboard, define stream or batch processing that you need
- **Data Visualization:** use D3.js, ApacheSolr + BANANA, Elasticsearch + KIBANA, or whatever you want to visualize the results obtained by analyzing your data log
- **Note:** *you may use thirdparty code (github, etc.) for implementing the activity recognition is not important complexity or accuracy of the task*



Temperature and Humidity in the classroom



Classroom Temperature and Humidity



- Let's try to use collect data from router for implementing things like, parental control or simply a network traffic analytics of my LAN and show them into an interactive dashboard
- **Data Ingestion:** collect data by using Arduino and MQTT + Mosquitto Broker
- **Data Storage:** choose and use the data file format, choose the repository (HDFS, KeyValue store, and so forth)
- **Data Processing:** Percentage of humidity in the environment; Heat Index is calculated; etc.
- **Data Visualization:** use D3.js, ApacheSolr + BANANA, Elasticsearch + KIBANA, or whatever you want to visualize the results obtained by analyzing your data log
- **Note:** I will give you some ESP32 and DHT11 sensors, or other sensors among the available ones <https://create.arduino.cc/projecthub/Gersaibot/temperature-streaming-with-arduino-big-data-tools-eb22fc>.



Raspberry Router



- Let's try to use collect data from router for implementing things like, parental control or simply a network traffic analytics of my LAN and show them into an interactive dashboard
- **Data Ingestion:** collect data about IPs, content topics, etc., considering the time dimension
- **Data Storage:** choose and use the data file format, choose the repository (HDFS, KeyValue store, and so forth)
- **Data Processing:** analytics question about network activities, pre-compute the results and then show them in a live dashboard, define stream or batch processing that you need
- **Data Visualization:** use D3.js, ApacheSolr + BANANA, Elasticsearch + KIBANA, or whatever you want to visualize the results obtained by analyzing your data log
- **Note:** I will give you raspberry, if needed, then you can collect data from it by configuring smart gateway following this guide <http://rbnet.it/guide/parental-control-con-raspberry-pi-squid-squidguard-e-dnsmasq/>



- Let's collect data from web or social media, and try to implement some interesting service to discover relevant information (OSIF)
- **Data Ingestion:** identify sources of information and what kind of resources you would like to manage, and for which objectives?
- **Data Storage:** choose and use the data file format, choose the repository (HDFS, KeyValue store, and so forth)
- **Data Processing:** use existing algorithms for pre-computing the results and then show them in a live dashboard.
- **Data Visualization:** use D3.js, ApacheSolr + BANANA, Elasticsearch + KIBANA, or whatever you want to visualize the results obtained by analyzing your data log
- **Note:** for instance let's find similar user profiles, or let's geolocate a selfie posted on social media, let's find if some selected users have same connection time on some network.



Facilities

Virtual Box



- Install Virtual Box: <https://www.virtualbox.org/wiki/Downloads>
- Download shared image:
<https://drive.google.com/open?id=113fCoZK2tYeWvyvkoUKRpYdQEtslbsOZ>
- Contents:
 - OS: Ubuntu
 - Pig
 - Hive
 - Neo4j
 - HDFS
 - PySpark
 - Jupyter Notebook

Virtual Machine



- Download VM from:
 - <https://drive.google.com/open?id=1ZCC5skAJSaAi9LbEEI8CKMWT7hBZqmz1>
- You need to install Oracle VirtualBox and then open the shared .ova file that contains the Ubuntu installation with some software already installed.

- Suggested video:[ITA] <https://www.youtube.com/watch?v=kYdP0BIGQAM>
- Visit: <https://www.docker.com/get-started>
- Download: Docker Desktop
- Let's start with docker next week for using HDFS and MapReduce
 - <https://github.com/effereds/hadoop>
 - From your terminal: `docker pull efferre/hadoop`